

m-format constellation -- Command-line Examples

mnemonic-toolkit v0.85.0 -- worked examples (Linux), exact verbatim I/O

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About these examples

This document shows real, copy-pasteable command lines for the **m-format constellation** -- a steel-engravable Bitcoin backup system built around four CLIs (`mnemonic`, `md`, `ms`, `mk`). Every command below was executed against `mnemonic v0.85.0` on Linux and **both the command and its full output are reproduced verbatim** -- no abbreviations, no ellipses, no elided keys or addresses. Long lines wrap with a grey hook-arrow continuation marker in the left margin.

All seed phrases here are public BIP-39 TEST VECTORS (`abandon abandon ... about`, etc.). They are world-known and hold no funds. **Never type a real seed phrase onto a networked machine, and never reuse these test phrases for real money.** The toolkit is alpha software -- use only with disposable amounts or on testnet until it has been independently audited.

Seed input from a file

Yes -- the toolkit reads a seed phrase from a file. There is no `--phrase-file` flag; instead every secret slot accepts the value `-`, meaning **"read this secret from stdin"**, which you point at a file with the shell's `< file` redirect:

```
$ mnemonic bundle --template bip84 --network mainnet --slot @0.phrase=- < seed.txt
```

This is the **secure** idiom: the phrase travels on stdin, so it never appears in `argv` / `/proc/$PID/cmdline` or your shell history (unlike the inline form `--slot @0.phrase='<words>'`, which the toolkit flags as `secret-on-argv`). The reader strips a single trailing newline, so an ordinary one-line text file works.

Multiple files for multiple seeds

Only **one** stdin secret is allowed per invocation, so `< file` reads exactly one seed. To use several seed files there are two patterns:

1. **One file per invocation (secure -- recommended).** Process each seed file in its own command and combine only the resulting public xpubs. No machine ever holds more than one seed, and nothing secret reaches `argv`. This is the per-device 2-of-3 multisig flow in section 3.
2. **Several files in one command (convenient -- less safe).** Read each file with command substitution, `--slot "@N.phrase=$(cat seedN.txt)"`. This works for any number of seeds, but each substituted phrase lands on `argv` (the toolkit prints a `secret-on-argv` warning). Only one slot may use the `-` stdin form; a second `-` is rejected. Shown in section 3.4.

Convention in this document: whenever a command reads a file (a seed, a descriptor, a policy JSON, or an `md1` chunk list), the file's contents are printed with `cat` immediately beforehand, so every input is visible.

The three engraved cards:

Card	What it carries
ms1	BIP-39 entropy (recovers the seed)
mk1	xpub + origin (master fingerprint + BIP path)
md1	wallet policy (descriptor template + bound xpubs)

Throughout, `$` is the shell prompt; everything after it is what you type.

1. Install the constellation on Linux

The in-repo installer builds each component with `cargo install --locked into ~/.cargo/bin` (no sudo, no system files touched). It needs cargo, git, and a C toolchain; the CLIs require `rustc >= 1.85`.

Install all four CLIs (this compiles from source, so the build log is machine-specific and not reproduced here):

```
$ sh -c "$(curl -fsSL https://raw.githubusercontent.com/bg002h/mnemonic-toolkit/master/scripts/install.sh)" -- --no-gui
```

The installer carries the current version pins, so it never goes stale. Useful flags: `--only <c>`, `--exclude <c>`, `--no-gui`, `--from-git`, `--force`, `--dry-run`, `--list`. The pin table (`--list`) and a dry run are deterministic (`$REPO` = your clone root):

```
$ sh "$REPO/scripts/install.sh" --list
COMPONENT      CARGO_PACKAGE  DEFAULT  FEATURES  GIT_TAG
-----
mnemonic        mnemonic-toolkit  git (only)  (none)    mnemonic-toolkit-v0.85.0
md              md-cli          crates.io  cli-compiler  descriptor-mnemonic-md-cli-v0.11.2
ms              ms-cli          crates.io  (none)     ms-cli-v0.14.1
mk              mk-cli          crates.io  (none)     mk-cli-v0.12.0
mnemonic-gui    mnemonic-gui     git (only)  (none)     mnemonic-gui-v0.51.0
```

```
$ sh "$REPO/scripts/install.sh" --no-gui --dry-run
m-format constellation installer
install root: /home/user/.cargo/bin
source: crates.io (default; mnemonic-toolkit stays on git+tag)

install mnemonic (git: mnemonic-toolkit-v0.85.0)
[dry-run] cargo install --locked --git https://github.com/bg002h/mnemonic-toolkit --tag mnemonic-toolkit-v0.85.0 mnemonic-toolkit
[dry-run] mkdir -p "/home/user/.local/share/man/man1" && "/home/user/.cargo/bin/mnemonic" gen-man --out "/home/user/.local/share/man/man1"
install md (crates.io: md-cli)
[dry-run] cargo install --locked --features cli-compiler md-cli
[dry-run] mkdir -p "/home/user/.local/share/man/man1" && "/home/user/.cargo/bin/md" gen-man --out "/home/user/.local/share/man/man1"
install ms (crates.io: ms-cli)
[dry-run] cargo install --locked ms-cli
[dry-run] mkdir -p "/home/user/.local/share/man/man1" && "/home/user/.cargo/bin/ms" gen-man --out "/home/user/.local/share/man/man1"
install mk (crates.io: mk-cli)
[dry-run] cargo install --locked mk-cli
[dry-run] mkdir -p "/home/user/.local/share/man/man1" && "/home/user/.cargo/bin/mk" gen-man --out "/home/user/.local/share/man/man1"
skip mnemonic-gui

4 installed.

verify:
  mnemonic --version      md --version
  ms --version            mk --version
  mnemonic-gui --version

man pages installed to /home/user/.local/share/man/man1;
if "man <cli>" does not find them, run: man -M "/home/user/.local/share/man/man1" <cli>
```

Verify the install and list every subcommand:

```
$ mnemonic --version
mnemonic 0.85.0
```

```
$ mnemonic --help
engraving-bundle CLI for the m-format star (msl + mkl + mdl)

Usage: mnemonic [OPTIONS] <COMMAND>

Commands:
  bundle          emit a 3-card engraving bundle from a phrase or xpub
  verify-bundle   round-trip-check an engraved bundle
  convert         convert between seed/key formats (BIP-39 / BIP-32 / WIF / msl / mkl)
  addresses       list a wallet's receive/change addresses (batch, read-only)
  decode-address  decode a Bitcoin address - network(s) / script type / witness version / scriptPubKey
  export-wallet   emit watch-only wallet artifacts (Bitcoin Core importdescriptors, BIP-388 wallet_policy)
  import-wallet   import a third-party wallet blob into an m-format bundle (v0.26.0 Phase 2: BSMS Round-2 only)
  derive-child    derive deterministic child entropy / keys from a master xprv (BIP-85)
  electrum-decrypt decrypt an Electrum field-encrypted secret (seed phrase / xprv) with a password
  final-word      emit the set of BIP-39 last words that yield a valid checksum for an N-1 partial phrase
  seed-xor       split a BIP-39 phrase into N XOR shares OR combine N shares back into a phrase
  seedqr         encode/decode SeedQR (BIP-39 mnemonic ↔ numeric digit-string QR payload)
  nostr           Wrap an existing nostr key (npub/nsec) as Bitcoin addresses/descriptors/WIF
  silent-payment Derive a BIP-352 silent-payment receiver address (base + labeled) from a seed
  slip39         split a master secret into SLIP-39 K-of-N shares OR combine shares back (Trezor-compatible)
  ms-shares      split a secret into BIP-93 codex32 K-of-N (msl) shares OR combine shares back
  gen-man        emit roff man pages for the whole CLI tree into a directory (clap-faithful)
  gui-schema     emit SPEC $7 GUI-overlay flag-surface schema JSON (companion to `mnemonic-gui` v0.2)
```

repair	BCH error-correct a corrupted m-format card (msl / mkl / mdl)
inspect	describe the contents of an m-format card (msl / mkl / mdl)
compare-cost	compare wsh-vs-tr per-spending-condition cost for a miniscript or descriptor
xpub-search	search for a target (xpub, descriptor, address, or passphrase) under a seed or xpub
verify-message	verify a Bitcoin message signature (legacy P2PKH signmessage + BIP-322 segwit/taproot)
restore	emit a watch-only restore document (single-sig) from a seed + optional passphrase
build-descriptor	build a validated wsh(...) descriptor + BIP-388 policy from a JSON policy-tree spec
word-card	encode an mkl/mdl card as an engravable BIP-39 Word Card (+ optional RAID), or --decode one back
help	Print this message or the help of the given subcommand(s)

Options:

```
--no-auto-repair v0.22.0 -- skip auto-fire repair on decode failures; preserve pre-v0.22 exit policy. Global flag. Honored by `convert`, `inspect`,
↳ and (v0.22.1+) `verify-bundle`. For `verify-bundle`, auto-fire is additionally gated on `std::io::stdout().is_terminal()` to preserve the
↳ legacy VerifyCheck-row behavior when output is piped or captured (per v0.22.1 D18 – TTY-conditional default). Standalone `repair` ignores this
↳ flag (the whole point of that subcommand IS repair). Under `--json` calling contexts the auto-fire emits a structured JSON envelope on stdout (
↳ per v0.22.1 D20) instead of text-form
-h, --help          Print help
-V, --version       Print version
```

RECOVERING A FORGOTTEN BIP-39 PASSPHRASE:

If you have your seed words (entropy) but not the BIP-39 passphrase (the optional "25th word"): if you have a LIST of likely passphrases, `mnemonic xpub-search passphrase-of-xpub --passphrase-candidates-file <file> --target-xpub <a known xpub>` tests each candidate against a value you already know. To GENERATE or mutate a keyspace (wordlists, masks, typo models), `mnemonic` does not – an external open-source tool does: btcrecover searches passphrase candidates and confirms each by deriving an address / xpub / master-fingerprint at common default paths and matching a value you already know.

btcrecover (maintained): <https://github.com/3rdIteration/btcrecover>
original: <https://github.com/gurnec/btcrecover>

Pointer current as of 2026-05-25. Run untrusted recovery tools offline, on an air-gapped machine.

2. Single-sig card set from a seed phrase (file input)

Create a native-segwit (BIP-84, $m/84'/0'/0'$) single-sig 3-card bundle from one seed phrase held in a file. Write the phrase to `seed0.txt` (here a public test vector) and feed it on stdin:

```
$ printf '%s\n' 'abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about' > seed0.txt
```

```
$ cat seed0.txt
abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about
```

(`--template` choices for single-sig: `bip44`, `bip49`, `bip84`, `bip86`.) Run the bundle. `stdout` carries the three cards to engrave; `stderr` carries the human-readable engraving panel and the secret-material warning:

```
$ mnemonic bundle --template bip84 --network mainnet --slot @0.phrase=- < seed0.txt
# ms1 (entropy, BCH-checksummed)
ms10e ntrsqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqj9 sxraq 34v7f

# mk1 (xpub + origin)
mklqp rsqhp qqsq3 cqtsl eeutk s2qvz g3vs7 0mej h622w s2kgd emj2c d8zvj 2skzx 2wq0q w70l4 q99vd yh5x0 z8v4y slsp8 qp3yx g3dpe 854wq 4
mklqp rsqhp p0f30 mtzxd 65mvw cur9u sdatw uqvq6 z70r9 nwrqk 6xn6l 8gy6n wa2n9 77sw6 zh34r ma0nh

# md1 (wallet policy)
mdlfg dxlpq pqp6 jzzq vqpdq w0za5 zs4gy y55aq 4vsmn hy4s6 wyayp u34c7 raqu8 np
mdlfg dxlpq f2zcg efcpu pme17 5q543 5j7se ugaj5 jr7qy ur6vt 76es5 cdeyr q7zdy 0d
mdlfg dxlpq 3xa2d k8vwp j7gx7 4hwqx qdp08 3jehp 5tdrf a0n5z dfkqc dlrnv h5r62 jn

# === Wallet bundle: bip84, mainnet ===
# ms1: 1c017
# mk1: 1c017
# fingerprint: 73c5da0a
# origin path: m/84'/0'/0'
# Template: bip84
# md1: 1c01
warning: stdout carries private key material (can spend) – redirect or encrypt (e.g. '> file.txt' or '| age -e ...')
```

Each card is printed once, grouped into 5-character blocks (`ms10e ntrsqqqqqq ...`) -- exactly the form you punch or engrave. Add `--no-engraving-card` to suppress the `stderr` panel when piping into other tools.

3. Conventional 2-of-3 multisig from 3 seed phrases (per-device, file input)

A real multisig never lets one machine see more than one seed. Each cosigner derives **only their public xpub** from their own seed file (on their own, ideally air-gapped, device); the coordinator then combines the three **public** keys into a watch-only `wsh(sortedmulti(...))`. No secret ever leaves its file.

Put each cosigner's seed in its own file:

```
$ printf '%s\n' 'abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about' > seed0.txt
```

```
$ printf '%s\n' 'legal winner thank year wave sausage worth useful legal winner thank yellow' > seed1.txt
```

```
$ printf '%s\n' 'letter advice cage absurd amount doctor acoustic avoid letter advice cage above' > seed2.txt
```

On each device, derive that cosigner's BIP-87 multisig fingerprint and account xpub (`--template wsh-sortedmulti` implies the `m/87'/0'/0'` path) from the seed file. Cosigner @0:

```
$ cat seed0.txt
abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about
```

```
$ mnemonic convert --from phrase=- --to fingerprint --template wsh-sortedmulti --network mainnet < seed0.txt
fingerprint: 73c5da0a
```

```
$ mnemonic convert --from phrase=- --to xpub --template wsh-sortedmulti --network mainnet < seed0.txt
xpub: xpub6DBjiYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuCG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM
note: stdout is watch-only – public keys only, cannot spend
```

Cosigner @1:

```
$ cat seed1.txt
legal winner thank year wave sausage worth useful legal winner thank yellow
```

```
$ mnemonic convert --from phrase=- --to fingerprint --template wsh-sortedmulti --network mainnet < seed1.txt
fingerprint: b8688df1
```

```
$ mnemonic convert --from phrase=- --to xpub --template wsh-sortedmulti --network mainnet < seed1.txt
xpub: xpub6CbhrPzY2z7NcCGCGjLAJLq8iRyjUfwmDXQs66MxTVURekqb9DpLnVJ5D1qpatZjUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokyLVsyEFZFt9
note: stdout is watch-only – public keys only, cannot spend
```

Cosigner @2:

```
$ cat seed2.txt
letter advice cage absurd amount doctor acoustic avoid letter advice cage above
```

```
$ mnemonic convert --from phrase=- --to fingerprint --template wsh-sortedmulti --network mainnet < seed2.txt
fingerprint: 28645006
```

```
$ mnemonic convert --from phrase=- --to xpub --template wsh-sortedmulti --network mainnet < seed2.txt
xpub: xpub6DB7HNqW6CZojxN85NuFTPWZhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
note: stdout is watch-only – public keys only, cannot spend
```

Wrap each as an origin-annotated descriptor key `[fingerprint/87'/0'/0']xpub` and combine into a 2-of-3 sorted-multisig descriptor (`/<0;1>/*` = the external/change multipath). The assembled descriptor file:

```
$ cat multisig.desc
wsh(sortedmulti(2,[73c5da0a/87'/0'/0']xpub6DBjiYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuCG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM
↳ /<0;1>/*,[b8688df1/87'/0'/0']xpub6CbhrPzY2z7NcCGCGjLAJLq8iRyjUfwmDXQs66MxTVURekqb9DpLnVJ5D1qpatZjUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokyLVsyEFZFt9
↳ /<0;1>/*,[28645006/87'/0'/0']xpub6DB7HNqW6CZojxN85NuFTPWZhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
↳ /<0;1>/*))
```

Validate and canonicalise it (this also computes the BIP-380 checksum):

```
$ mnemonic export-wallet --descriptor "$(cat multisig.desc)" --format descriptor --network mainnet
wsh(sortedmulti(2,[73c5da0a/87'/0'/0']xpub6DBjYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuCG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM
↳ /<0;1>/*,[b8688df1/87'/0'/0']xpub6CbhrPzY2z7NcCGGjLAJLq8iRyJufwmdXQs66MxTVURekqb9DpLnVJ5D1qpatZjUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokyLVsyEFZFt9
↳ /<0;1>/*,[28645006/87'/0'/0']xpub6DB7HNqw6CZojxN85NuFTPWZhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
↳ /<0;1>/*) #4wup4at0
note: stdout is watch-only – public keys only, cannot spend
```

The first receive address (here via the BSMS / BIP-129 record, which also carries the /0*/,/1/* derivation):

```
$ mnemonic export-wallet --descriptor "$(cat multisig.desc)" --format bsms --network mainnet
BSMS 1.0
wsh(sortedmulti(2,[73c5da0a/87'/0'/0']xpub6DBjYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuCG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM
↳ /<0;1>/*,[b8688df1/87'/0'/0']xpub6CbhrPzY2z7NcCGGjLAJLq8iRyJufwmdXQs66MxTVURekqb9DpLnVJ5D1qpatZjUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokyLVsyEFZFt9
↳ /<0;1>/*,[28645006/87'/0'/0']xpub6DB7HNqw6CZojxN85NuFTPWZhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
↳ /<0;1>/*) #4wup4at0
/0*/,/1/*
bclqkssenl2m6t3aynza394sr9m86vt6md2v76kj52jun2xlrdeaa4q84qtpl
note: stdout is watch-only – public keys only, cannot spend
```

Engrave the shared watch-only card set from the public descriptor (the md1 policy card is shared by all cosigners; each cosigner additionally backs up their own seed as a single-sig ms1 set per section 2). With only public xpubs supplied, the ms1 cards are empty placeholders:

```
$ mnemonic bundle --descriptor-file multisig.desc --network mainnet
# ms1 (omitted – descriptor watch-only mode)

# mk1[0] (cosigner 0 xpub + origin)
mklqp erpup qqspu 3s7de nyv8n wverp umnrr chdq5 pcy3z epa59 349cd gs5n5 stvpv k3v8x 4eqsd ngy2j l4wdd t5ac2 ptv4f ya76t preap yrdfq r
mklqp erpup pnq8k p6xcp qphr7 svxwk xx5ag 99s9z fyml9 v7tcq rexmp dj7jg qgmny 8rr0z 7vlj7 eqzv2 486xk kcftr zq8

# mk1[1] (cosigner 1 xpub + origin)
mklqp erpup qqspu 3s7de nyv8n wverp umn9c dzx1z pcy3z epa9x afl63 0m4as 45q6f z4lts ntlue s3e3g ylcu6 jsa6j dz69h y0whc pg2f2 8j5aw u
mklqp erpup p02p3 qtfjg qpxtu j95jz aevqs 7jqje 40324 vxlfw 0txsw awpxt e3zmh zp6lr pj3ga 2lw65 h2dd2 4cuer kpa

# mk1[2] (cosigner 2 xpub + origin)
mklqp erp7p qqspu 3s7de nyv8n wverp umnpg v3gqv pcy3z epa9e cw97s n9g25 uukrg xca47 at8as 54xhd 0rkxg 57cs7 8mc8f k6507 cs8tn ugnpn f
mklqp erp7p pr4e5 9ef7u 5p038 j900y e439h sphez 506fr a5k3e ac5kg 0n7jh n3kpz pna6n 0lygd lpfeq 4a75l c2mrr z7f

# md1 (multisig wallet policy)
md1f5 przsz pq3m6 7zzqz vzrs3 pstuc w0za5 znwrg 3hcc5 xg5qx zhs7y yg2f6 g9kqk tgrkrn 2spp6 tlfzp rv6ye
md1f5 przsz wgyrv 6p25h atnt2 a8wzs 2m92f fsrmq arvqs qm3lg xr8tr r2w5z jcz3y jdljk 0qf2g 9tn3u 5jtsz
md1f5 przsz j7q77 fkcvt h5jqz xuepc cmchn 8u30m 4as45 q6fz4 ltsnt lues3 e3gyl cu6js asrwq tqpe4 mvjxf
md1f5 przsz afx3d zmj02 p3qtf fjqpj tuj95 jzaev qs7jq je403 24vxl fw0tx swawp xte3z mspu3 5hx7l rg2t6
md1f5 przsz r3qa0 3segk px2s4 feevx sd3mt a6k0m pf2dw 678vs dfa3p u0hsw nvwhx sh98m jsqps z8xkj h9l6g
md1f5 przsz gzl20 y277f ntzt0 qr0j9 gl5j8 mfdrr m3fvs l8a90 8rvzy r8m4x l7gs9 e2t6d yvjgv 4j

# === Wallet bundle: descriptor, mainnet ===
# Threshold: 2 of 3
# Cosigners:
# @0: (no ms1; watch-only),mk1:c8c3c (73c5da0a @ m/87'/0'/0')
# @1: (no ms1; watch-only),mk1:c8c3d (b8688df1 @ m/87'/0'/0')
# @2: (no ms1; watch-only),mk1:c8c3e (28645006 @ m/87'/0'/0')
# Template: descriptor
# md1: c8c3
# Recovery: any 2 of 3 signing keys + md1 (template card).
note: stdout is watch-only – public keys only, cannot spend
```

3.4 Building from all seeds on one machine (multiple files, one command)

If instead you hold all the seeds yourself, you can build the whole bundle in a single command, reading each seed from its own file with command substitution. This is less safe -- each substituted phrase lands on `argv`, so the toolkit prints a secret-on-argv warning for every one -- but it needs no per-device coordination. The three seed files (shown again for reference):

```
$ cat seed0.txt
abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about
```

```
$ cat seed1.txt
legal winner thank year wave sausage worth useful legal winner thank yellow
```

```
$ cat seed2.txt
letter advice cage absurd amount doctor acoustic avoid letter advice cage above
```

Because seeds (not just xpubs) are supplied, this emits the **full secret card set** -- one `ms1` per cosigner -- not the watch-only placeholders of 3.3:

```
$ mnemonic bundle --template wsh-sortedmulti --threshold 2 --network mainnet --slot "@0.phrase=$(cat seed0.txt)" --slot "@1.phrase=$(cat seed1.txt)" --
↳ slot "@2.phrase=$(cat seed2.txt)"
warning: secret material on argv (--slot @0.phrase=) - pipe via --slot @0.phrase= to avoid /proc/$PID/cmdline exposure
warning: secret material on argv (--slot @1.phrase=) - pipe via --slot @1.phrase= to avoid /proc/$PID/cmdline exposure
warning: secret material on argv (--slot @2.phrase=) - pipe via --slot @2.phrase= to avoid /proc/$PID/cmdline exposure
# ms1[0] (entropy, BCH-checksummed)
ms10e ntrsq qqqqq qqqqq qqqqq qqqqq qqccj9 sxraq 34v7f

# ms1[1] (entropy, BCH-checksummed)
ms10e ntrsq plh7l ml0al h7lml 0alh7 lml0a ls5cc lar2z mksh6

# ms1[2] (entropy, BCH-checksummed)
ms10e ntrsq zqgpq yqszy gpqyq szqgp qyqszy qqlfm 7mep8 4hunu

# mk1[0] (cosigner 0 xpub + origin)
mk1qp erpup qqspu 3s7de nyv8n wverp umnrn chdq5 pcy3z epa59 349dc gs5n5 stvpv k3v8x 4eqsd ngy2j l4wdd t5ac2 ptv4f ya76t preap yrdqf r
mk1qp erpup pnq8k p6xcp qphr7 svxwk xx5ag 99s9z fym19 v7tcq rexp dj7jg qgmny 8rr0z 7vlj7 eqzv2 486xk kcftr zq8

# mk1[1] (cosigner 1 xpub + origin)
mk1qp erpap qqspu 3s7de nyv8n wverp umn9c dzxlz pcy3z epa9x afl63 0m4as 45q6f z4lts ntlue s3e3g ylcu6 jsa6j dz69h y0whc pg2f2 8j5aw u
mk1qp erpap p02p3 qtffj qpxtu j95jz aevqs 7jqje 40324 vxlfw 0txsw awpzt e3zmh zp6lr pj3ga 2lw65 h2dd2 4cuer kpa

# mk1[2] (cosigner 2 xpub + origin)
mk1qp erp7p qqspu 3s7de nyv8n wverp umnpg v3gqv pcy3z epa9e cw97s n9g25 uukrg xca47 at8as 54xhd 0rk9x 57cs7 8mc8f k6507 cs8tn ugnpn f
mk1qp erp7p pr4e5 9ef7u 5p038 j900y e439h sphez 506fr a5k3e ac5kg 0n7jh n3kpz pna6n 0lygd lpfeq 4a75l c2mrr z7f

# md1 (multisig wallet policy)
md1f5 przsz pq3m6 7zzqz vzrs3 pstuc w0za5 znwrg 3hcc5 xg5qx zhs7y yg2f6 g9kqk tgkrn 2spp6 tlfzp rv6ye
md1f5 przsz wgyrv 6pz5h atnt2 a8wz5 2m92f f5rmq arvqs qm3lg xr8tr r2w5z jc23y jdljk 0qf2g 9tn3u 5jtsz
md1f5 przsz j7qq7 fkctv h5jqz xuepc cmchn 8u30m 4as45 q6fz4 ltsnt lues3 e3gyl cu6js asrwq tqpe4 mvjxf
md1f5 przsz afx3d zmj02 p3qtf fjqpz tuj95 jzaev qs7jq je403 24vxl fw0tx swapx xte3z mspu3 5hx7l rg2t6
md1f5 przsz r3qa0 3segk px2s4 feevx sd3mt a6k0m pf2dw 678vs dfa3p u0hsw nvwhx sh98m jsqps z8xkj h9l6g
md1f5 przsz gzlz0 y277f ntzt0 qr0j9 gl5j8 mfdn m3fvs l8a90 8rvzy r8m4x l7gs9 e2t6d yvjgv 4j

# === Wallet bundle: wsh-sortedmulti, mainnet ===
# Threshold: 2 of 3
# Cosigners:
# @0: ms1:c8c3c,mk1:c8c3c (73c5da0a @ m/87'/0'/0')
# @1: ms1:c8c3d,mk1:c8c3d (b8688dfl @ m/87'/0'/0')
# @2: ms1:c8c3e,mk1:c8c3e (28645006 @ m/87'/0'/0')
# Template: wsh-sortedmulti
# md1: c8c3
# Recovery: any 2 of 3 signing keys + md1 (template card).
warning: stdout carries private key material (can spend) - redirect or encrypt (e.g. '> file.txt' or '| age -e ...')
```

Only one secret may arrive on stdin, so you cannot replace more than one substitution with the `=` file-redirect form -- a second `=` is rejected:

```
$ mnemonic bundle --template wsh-sortedmulti --threshold 2 --network mainnet --slot @0.phrase= --slot @1.phrase= --slot "@2.phrase=$(cat seed2.txt)" <
↳ seed0.txt
warning: secret material on argv (--slot @2.phrase=) - pipe via --slot @2.phrase= to avoid /proc/$PID/cmdline exposure
error: at most one --slot @N.<secret>= per invocation (single stdin per invocation)
```


4. Card set -> Bitcoin Core wallet descriptor (and how to import)

mnemonic restore --md1 <chunks> reconstructs the watch-only wallet from the **shared md1 card alone** -- no seeds needed. First produce that card from the section-3 wallet (descriptor file shown again) and pull out its md1 chunks:

```
$ cat multisig.desc
wsh(sortedmulti(2,[73c5da0a/87'/0'/0']xpub6DBj1Ync4ewKti13Q1L35bqodow5z3VGJnf516B3icHrEGEUcCuG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqrXvicM
  ↳ /<0;1>/*,[b8688df1/87'/0'/0']xpub6CbhRzPzY2z7NcCGGjLAJLQ8iRyjUfwmDX0s66MxTVUReKqB9DpLnVJ5D1qpatZjUuPGTyxf5TYUlvA34YFE9FHB4TvfyMokYLVsyEFZft9
  ↳ /<0;1>/*,[28645006/87'/0'/0']xpub6DB7HNQw6CZoJxN85NuFTPWhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTfMFKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
  ↳ /<0;1>/*))
```

```
$ mnemonic bundle --descriptor-file multisig.desc --network mainnet --json | jq -r ".md1[]" > multisig.md1
note: stdout is watch-only - public keys only, cannot spend
```

```
$ cat multisig.md1
md1f5przzspq3m67zzqzvrs3pstucw0za5znwrg3hcc5xg5qxzhs7y9g2f6g9kqktgkrn2spp6tlfzprv6ye
md1f5przzswgyrv6pz5hatnt2a8wzs2m92ffsrmqarvqsm3lgxr8trr2w5zjcz3yjdLjk0qf2g9tn3u5jtsz
md1f5przzsj7q7fkctvh5jqzxuepcmcn8u30m4as45q6fz4ltsntlues3e3gylcu6j5asrwtqpe4mvjxf
md1f5przzsafx3dzmj02p3qtfjqpxtuj95jzaevqs7jqje40324vxlfw0txswawpxte3zmspu35hx7lrg2t6
md1f5przz3rq03se9kpx2s4feevsd3mta6k0mpf2dw678vdsfa3pu0hswvwhxsh98mjsqps28xkjh9l6g
md1f5przz3gzl0y277fntzt0qr0j9gl5j8mfdnm3fvs18a908rvzyr8m4xl7gs9e2t6dyvjgv4j
```

Restore reconstructs the wallet from exactly those chunks. The default form prints the descriptor and first address (note the address matches section 3 -- same wallet -- while the descriptor *string* differs because the md1 card stores each key as a depth-0 master xpub, an equivalent serialisation):

```
$ mnemonic restore --network mainnet $(sed "s/~/--md1 /" multisig.md1)
2-of-3 multisig restore
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: wsh(sortedmulti(2,[73c5da0a/87'/0'/0']
  ↳ xpub661MyMwAQBcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzw8TBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPWDUSNJsY5JsgVBQoD6KrbdCLL/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub661MyMwAQBcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXyukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmREkfhjzgHCUEcvJBhXS5bk4RW/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub661MyMwAQBcEddy4jr5EtEHQBctfscE6a99DGLr2cW4HnnmBSXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/<0;1>/*))#yjp7hj7w
first recv: bc1qkssen12m6t3aynza394sr9m86vt6md2v76kj52jun2x1wrdeaa4q84qtpl
cosigner @0: 73c5da0a [87'/0'/0'] from md1 (not independently verified)
cosigner @1: b8688df1 [87'/0'/0'] from md1 (not independently verified)
cosigner @2: 28645006 [87'/0'/0'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only - public keys only, cannot spend
```

Add --format bitcoin-core for a ready-to-import importdescriptors request array (external .../0/* + change .../1/*):

```
$ mnemonic restore --network mainnet $(sed "s/~/--md1 /" multisig.md1) --format bitcoin-core
[
  {
    "active": true,
    "desc": "wsh(sortedmulti(2,[73c5da0a/87'/0'/0']
      ↳ xpub661MyMwAQBcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzw8TBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPWDUSNJsY5JsgVBQoD6KrbdCLL/0/*,[b8688df1/87'/0'/0']
      ↳ xpub661MyMwAQBcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXyukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmREkfhjzgHCUEcvJBhXS5bk4RW/0/*,[28645006/87'/0'/0']
      ↳ xpub661MyMwAQBcEddy4jr5EtEHQBctfscE6a99DGLr2cW4HnnmBSXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/0/*))#y65a0dtg",
    "internal": false,
    "range": [
      0,
      999
    ],
    "timestamp": 0
  },
  {
    "active": true,
    "desc": "wsh(sortedmulti(2,[73c5da0a/87'/0'/0']
      ↳ xpub661MyMwAQBcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzw8TBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPWDUSNJsY5JsgVBQoD6KrbdCLL/1/*,[b8688df1/87'/0'/0']
      ↳ xpub661MyMwAQBcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXyukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmREkfhjzgHCUEcvJBhXS5bk4RW/1/*,[28645006/87'/0'/0']
      ↳ xpub661MyMwAQBcEddy4jr5EtEHQBctfscE6a99DGLr2cW4HnnmBSXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/1/*))#k0gfvz2t",
    "internal": true,
    "range": [
      0,
      999
    ],
    "timestamp": 0
  }
]
2-of-3 multisig restore
CONFIRM: verify each cosigner fingerprint against your records before importing the payload above.
descriptor: wsh(sortedmulti(2,[73c5da0a/87'/0'/0']
  ↳ xpub661MyMwAQBcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzw8TBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPWDUSNJsY5JsgVBQoD6KrbdCLL/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub661MyMwAQBcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXyukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmREkfhjzgHCUEcvJBhXS5bk4RW/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub661MyMwAQBcEddy4jr5EtEHQBctfscE6a99DGLr2cW4HnnmBSXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/<0;1>/*))#yjp7hj7w
first recv: bc1qkssen12m6t3aynza394sr9m86vt6md2v76kj52jun2x1wrdeaa4q84qtpl
```

```
cosigner @0: 73c5da0a [87'/0'/0'] from md1 (not independently verified)
cosigner @1: b8688df1 [87'/0'/0'] from md1 (not independently verified)
cosigner @2: 28645006 [87'/0'/0'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only – public keys only, cannot spend
```

Import into Bitcoin Core: save the array, create a blank descriptor wallet, and load it (these run against your own node, so their output is not shown here):

```
$ mnemonic restore --network mainnet $(sed "s/^/--md1 /" multisig.md1) --format bitcoin-core > wallet.json
$ bitcoin-cli -named createwallet wallet_name="multisig-2of3" disable_private_keys=true blank=true descriptors=true
$ bitcoin-cli -rpcwallet="multisig-2of3" importdescriptors "$(cat wallet.json)"
$ bitcoin-cli -rpcwallet="multisig-2of3" getnewaddress
```

Tips: `--timestamp now` skips the rescan for a fresh wallet (default 0 rescans from genesis); `--range 0,4999` widens the gap limit; `--bitcoin-core-version 24` targets older Core. `restore` also emits `--format descriptor` (the bare `wsh(...)#checksum`) for other wallets.

5. Custom degrading-miniscript wallet -- the pathological example (distinct keys per tier) + watch-only export

A four-tier vault -- our **pathological example** wallet. Each tier uses its own distinct key set (no key reuse), deliberately mixing all four Bitcoin timelock kinds:

Tier	Spend condition	Timelock kind
1	3-of-3 (K0,K1,K2) + secret word	absolute height -- after(1000000)
2	2-of-3 (K3,K4,K5) + secret word	absolute time -- after(1893456000)
3	both K6 and K7	relative blocks -- older(65535)
4	any 1 of K8,K9,K10	relative time -- older(4255898)

That is **11 distinct keys** (3+3+2+3). Absolute locks (after) count from the chain's height/clock; relative locks (older) count from each coin's own confirmation. Encodings:

- after(1000000) -- absolute **block height** 1,000,000 (BIP-65; values below 500,000,000 are heights).
- after(1893456000) -- absolute **Unix time** = 2030-01-01 00:00 UTC (values at/above 500,000,000 are timestamps).
- older(65535) -- relative **blocks**: 65,535 blocks (~455 days). This is the largest safe relative-block lock; older(65536) would be BIP-68 consensus-masked to zero, and the toolkit warns if it sees one.
- older(4255898) -- relative **time**: BIP-68's time flag (bit 22) set, plus 61,594 units x 512 s ≈ 365 days.

5.1 The secret word (a hashlock shared by tiers 1 and 2)

Reusing a *hash* across tiers is fine -- it is not a key. The secret word is `opensesame`; the descriptor commits to `H = sha256(⇨ sha256(word))` and a spend reveals the 32-byte preimage `x = sha256(word)`:

```
$ python3 -c "import hashlib; w='opensesame'; X=hashlib.sha256(w).digest(); print('preimage X =', X.hex()); print('hash H      =', hashlib.sha256(X).hexdigest())"
preimage X = cd6f70a4440de96063f8dbd7a4a3bbcff6b993af6c83e08d87b359e0760ca9c6
hash H      = a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad
```

5.2 The guided builder caps complexity -- use the raw descriptor path

`mnemonic build-descriptor` runs a satisfiability + cost preview that it **bounds** for funds-safety. An 11-key, 4-branch policy exceeds that envelope, so the guided builder refuses and points you at the raw `--descriptor` path. The policy-tree spec it reads:

```
$ cat policy.json
{
  "schema_version": 1,
  "wrapper": "wsh",
  "root": {
    "or_i": [
      {
        "and_v": [
          {
            "wrap": {
              "w": "v",
              "sub": {
                "after": 1000000
              }
            }
          },
          {
            "wrap": {
              "w": "v",
              "sub": {
                "sha256": "a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad"
              }
            }
          }
        ]
      },
      {
        "multi": {
          "k": 3,
          "keys": [
```

```

        "[73c5da0a/84'/0'/0']xpub6CatWdiZiodmUeTdp8LT5or8nmbKNcuyvz7WyksVFkKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLoP2fMFMKHPJ4ZeZXYVUhLv1VMrjPC7PW6V
    ↪ ",
    ↪ "[73c5da0a/84'/0'/1']xpub6CatWdiZiodmYVtWLtEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEZv3quZSJKa5dh5Lo2PbubMGxCfZtRthV6ST2qquL9w3HSzcUn
        "[73c5da0a/84'/0'/2']xpub6CatWdiZiodmbNGqcQxxjGN1650xTU4PwNNi9WrjYgYf7VxcmuFfosRw3focZLgRDbjDjJbqZhPCTfkcaWmL9BuSw98ybKKJtHcHgWeryy6"
    }
    }
    ]
    }
    ],
    {
    "or_i": [
    {
    "and_v": [
    {
    "wrap": {
    "w": "v",
    "sub": {
    "after": 1893456000
    }
    }
    },
    {
    "and_v": [
    {
    "wrap": {
    "w": "v",
    "sub": {
    "sha256": "a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad"
    }
    }
    },
    {
    "multi": {
    "k": 2,
    "keys": [
    "[73c5da0a/84'/0'/3']
    ↪ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsyEAaQfkjKwbwUi79LAWG5gHjMCQB7BeJc47MkubXuZdf45JZHK1qcr1GZ5EwREUDVDLVdPkEC",
    "[b8688df1/84'/0'/0']
    ↪ xpub6DNfJehqF1LU9s9kwaqDuI2Ajpz9psYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDujWHzVy1AtV6ENGKWgwbLWqa4wXMZR4ZmdpRjQB65EgTV",
    "[b8688df1/84'/0'/1']
    ↪ xpub6DNfJehqF1LUsoh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwY71Nc6LvqfDLWuyCivJ4eMDxgFtiTxnazgp8rYVzQSKR3L8EGRS9asBarqpe"
    ]
    }
    }
    ]
    }
    ],
    {
    "or_i": [
    {
    "and_v": [
    {
    "wrap": {
    "w": "v",
    "sub": {
    "older": 65535
    }
    }
    },
    {
    "multi": {
    "k": 2,
    "keys": [
    "[b8688df1/84'/0'/2']
    ↪ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqsHpQ6yNGdG4VXGKbLgcxKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51",
    "[b8688df1/84'/0'/3']
    ↪ xpub6DNfJehqF1LUxb8GRSstbR9LcAxWgwD4V678z5FZ7BLftWzwQCt3yZb5eW4U8AVJwbKcZ2ieggjvKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU"
    ]
    }
    }
    ]
    },
    {
    "and_v": [
    {
    "wrap": {
    "w": "v",
    "sub": {
    "older": 4255898
    }
    }
    }
    ],

```

```

{
  "multi": {
    "k": 1,
    "keys": [
      "[28645006/84'/0'/0']"
      ↪ xpub6DBbZvudcQg2nS4tHskrm7FGkXL6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuZTzJtwDveibGd83eGzdGCR6NjSqU8p2Xx4wWQK7iGBycAm",
      "[28645006/84'/0'/1']"
      ↪ xpub6DBbZvudcQg2sPDRWpqfEn5VzPiwrD1zNes4aHmNUog9Jmc2fc2J5fM2E39Yz2iakmpWRpa3rXhzGNmd4GKiJEZxmTCftQpwBGd9ihVks",
      "[28645006/84'/0'/2']"
      ↪ xpub6DBbZvudcQg2uk9BrxsuxCWjsYsbfPYkPahQfmTVABfJ4j8TRxUnRd5eGgEXfgPJ63xcuTV9uny7pQBFbu3XKcN9rNxcNGRaDT9BD1gkBGZ"
    ]
  }
}

```

Running the guided builder on it:

```

$ mnemonic build-descriptor --spec policy.json --network mainnet
build-descriptor: refused - 1 diagnostic(s):
  [over_envelope] root: policy exceeds the always-previewable envelope (2^(11 keys + 2 hashes) × 9 timelock-states > cap 4096); use the raw '--descriptor
  ↪ ` path for arbitrarily complex policies

```

(For a policy *within* the envelope -- fewer keys -- build-descriptor --spec validates and emits it for you.) For arbitrarily complex policies you hand the miniscript descriptor straight to export-wallet / bundle. The hand-written descriptor file:

```

$ cat policy.desc
wsh(or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,[73c5da0a/84'/0'/0']
  ↪ xpub6CatWdiZiodmUeTPd8LT5or8nmbKNcuYvz7WYksVFkKB4RHwCD3YxuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↪ xpub6CatWdiZiodmYVtWLTqEsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZ5JKA5dh5Lo2PbubMGxCFzTtrthV6ST2qquL9w3H5zcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↪ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNni9WrijYgYf7VxcmuFxsRw3focZLgRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgWeryy6/<0;1>/*),or_i(and_v(v:after
  ↪ (1893456000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(2,[73c5da0a/84'/0'/3']
  ↪ xpub6CatWdiZiodmHurRokjbycCrxdTDJgTsyEaAfKjKwbWUi79LAWG5gHjMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLvpKEC/<0;1>/*,[b688df1/84'/0'/0']
  ↪ xpub6DNfJehqF1LU59kwaqDu12Ajpz9psVYtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDujWhZVy1AtV6ENGKwgbLWqa4wXMZr4ZmdpRjQBG5EgTV/<0;1>/*,[b688df1/84'/0'/1']
  ↪ xpub6DNfJehqF1LUsoh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvkqfDLWuyCivJ4eMDxgftiTxnazgprYvZQSKR3L8EGRs9asBarqpe/<0;1>/*),or_i(and_v(v:older
  ↪ (65535),multi(2,[b688df1/84'/0'/2']
  ↪ xpub6DNfJehqF1LUwZ1DRfCQn365CDMPLSrqpSwaz4eRrtYz9qWdsVH9JqshP6yNgdG4VXGkLgxcKkXtWtB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b688df1/84'/0'/3']
  ↪ xpub6DNfJehqF1LUxb8gRSstbR9LcAxWgwD4V678z5FZ7BLftWzWqCt3yZb5eW4U8AVJwbKcZ2ieggjvKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*),and_v(v:older(4255898)
  ↪ ,multi(1,[28645006/84'/0'/0']xpub6DBbZvudcQg2nS4tHskrm7FGkXL6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuZTzJtwDveibGd83eGzdGCR6NjSqU8p2Xx4wWQK7iGBycAm
  ↪ /<0;1>/*,[28645006/84'/0'/1']xpub6DBbZvudcQg2sPDRWpqfEn5VzPiwrD1zNes4aHmNUog9Jmc2fc2J5fM2E39Yz2iakmpWRpa3rXhzGNmd4GKiJEZxmTCftQpwBGd9ihVks
  ↪ /<0;1>/*,[28645006/84'/0'/2']xpub6DBbZvudcQg2uk9BrxsuxCWjsYsbfPYkPahQfmTVABfJ4j8TRxUnRd5eGgEXfgPJ63xcuTV9uny7pQBFbu3XKcN9rNxcNGRaDT9BD1gkBGZ
  ↪ /<0;1>/*))#4ld0crrxa

```

Validate and canonicalise it (this adds the BIP-380 checksum). The full canonical descriptor, with every xpub in full:

```

$ mnemonic export-wallet --descriptor "$(cat policy.desc)" --format descriptor --network mainnet
wsh(or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,[73c5da0a/84'/0'/0']
  ↪ xpub6CatWdiZiodmUeTPd8LT5or8nmbKNcuYvz7WYksVFkKB4RHwCD3YxuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↪ xpub6CatWdiZiodmYVtWLTqEsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZ5JKA5dh5Lo2PbubMGxCFzTtrthV6ST2qquL9w3H5zcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↪ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNni9WrijYgYf7VxcmuFxsRw3focZLgRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgWeryy6/<0;1>/*),or_i(and_v(v:after
  ↪ (1893456000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(2,[73c5da0a/84'/0'/3']
  ↪ xpub6CatWdiZiodmHurRokjbycCrxdTDJgTsyEaAfKjKwbWUi79LAWG5gHjMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLvpKEC/<0;1>/*,[b688df1/84'/0'/0']
  ↪ xpub6DNfJehqF1LU59kwaqDu12Ajpz9psVYtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDujWhZVy1AtV6ENGKwgbLWqa4wXMZr4ZmdpRjQBG5EgTV/<0;1>/*,[b688df1/84'/0'/1']
  ↪ xpub6DNfJehqF1LUsoh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvkqfDLWuyCivJ4eMDxgftiTxnazgprYvZQSKR3L8EGRs9asBarqpe/<0;1>/*),or_i(and_v(v:older
  ↪ (65535),multi(2,[b688df1/84'/0'/2']
  ↪ xpub6DNfJehqF1LUwZ1DRfCQn365CDMPLSrqpSwaz4eRrtYz9qWdsVH9JqshP6yNgdG4VXGkLgxcKkXtWtB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b688df1/84'/0'/3']
  ↪ xpub6DNfJehqF1LUxb8gRSstbR9LcAxWgwD4V678z5FZ7BLftWzWqCt3yZb5eW4U8AVJwbKcZ2ieggjvKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*),and_v(v:older(4255898)
  ↪ ,multi(1,[28645006/84'/0'/0']xpub6DBbZvudcQg2nS4tHskrm7FGkXL6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuZTzJtwDveibGd83eGzdGCR6NjSqU8p2Xx4wWQK7iGBycAm
  ↪ /<0;1>/*,[28645006/84'/0'/1']xpub6DBbZvudcQg2sPDRWpqfEn5VzPiwrD1zNes4aHmNUog9Jmc2fc2J5fM2E39Yz2iakmpWRpa3rXhzGNmd4GKiJEZxmTCftQpwBGd9ihVks
  ↪ /<0;1>/*,[28645006/84'/0'/2']xpub6DBbZvudcQg2uk9BrxsuxCWjsYsbfPYkPahQfmTVABfJ4j8TRxUnRd5eGgEXfgPJ63xcuTV9uny7pQBFbu3XKcN9rNxcNGRaDT9BD1gkBGZ
  ↪ /<0;1>/*))#4ld0crrxa
note: stdout is watch-only - public keys only, cannot spend

```

First receive address (Mainnet), via the BSMS record:

```

$ mnemonic export-wallet --descriptor "$(cat policy.desc)" --format bsms --network mainnet
BSMS 1.0
wsh(or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,[73c5da0a/84'/0'/0']
  ↪ xpub6CatWdiZiodmUeTPd8LT5or8nmbKNcuYvz7WYksVFkKB4RHwCD3YxuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↪ xpub6CatWdiZiodmYVtWLTqEsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZ5JKA5dh5Lo2PbubMGxCFzTtrthV6ST2qquL9w3H5zcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↪ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNni9WrijYgYf7VxcmuFxsRw3focZLgRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgWeryy6/<0;1>/*),or_i(and_v(v:after
  ↪ (1893456000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(2,[73c5da0a/84'/0'/3']
  ↪ xpub6CatWdiZiodmHurRokjbycCrxdTDJgTsyEaAfKjKwbWUi79LAWG5gHjMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLvpKEC/<0;1>/*,[b688df1/84'/0'/0']

```

```

↳ xpub6DNfJehqF1LUs9kwaqDu12Ajzp9psYVtbGhTyKq0iCYdkkqV2vArY4diWxSTTDujWhzVy1AtV6ENGKgwblWqa4wXmZR4ZmdpRjQBG5EgTV/<0;1/*, [b8688df1/84/'0/'1']
↳ xpub6DNfJehqF1LUs0h1a4XLG3yFu1oCRjdB6Kba97xwV6SwY7I1nc6LvqkFDLWuyCivJ4MEdxgFtiXnazgpp8rYvZQSKR3L8EGrs9asBarqqe/<0;1/*))))),or_i(and_v(v:older
(65535),multi(2,[b8688df1/84/'0/'2']
↳ xpub6DNfJehqF1LUwZ1DrFCQ365CDMPLSrqpSwaz4eRrtYz9qWdsVH9J3sHqP06yNgdG4VXGKbLgkcKkXtwTB584iUcHmjomqR8z6NNxNpU51/<0;1/*, [b8688df1/84/'0/'3']
↳ xpub6DNfJehqF1LUx8bGR5stBR9LcAxhwgD4V678z5FZ7BLftwzWQcT3yZb5eW4U8AVJwbKcZ2EgijvkvXzggJpMBRkk3CE7b794uMV7Qp3TU/<0;1/*))))),and_v(v:older(4255898)
↳ ,multi(1,[28645006/84/'0/'0']xpub6DBBzvudcQgZnS4tHskrm7F6Kl6arWxEUz1g16kMaNdWRobCjmxH8KQdjuzTzJTdveibGd83GzGdGCR6NjsqUp8b2Xx4wWQK7i4B5yCa
/<0;1/*, [28645006/84/'0/'1']xpub6DBBzvudcQgZsPDRWpQfEn5VzPiwrd1Znes4AhhmNUog9Jmc2fC2J5FM2E39Yz2iaakmpWRpa3rXzhGmmd4GKijE2xmTcTfQpWbGd9i9hVks
/<0;1/*, [28645006/84/'0/'2']xpub6DBBzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfvmTABfJ4rJ8TRxUnDr5eGgEXfgPj63xcuTV9uny7pQBfbu3XKCN9rxNxcNGRaDT9BD1gkBGZ
/<0;1/*))))))#4ld0crxa
/0/*,/1/*
bc1qg7564xxd9jh68hquw5e558cqafhsklerkr0asfzqp6puq74veesrp6qss
note: stdout is watch-only – public keys only

```

5.3 Engrave the card set

Because every key is **distinct**, this is a valid BIP-388 wallet policy, so -- unlike a key-reusing policy -- `bundle` will engrave it. With only public xpubs supplied, the result is watch-only (the `ms1` cards are empty placeholders). The descriptor file it reads:

```
cat policy.desc
wsh(or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0flac62dbd829b9a08ad),multi(3,[73c5da0a/84/'0/'0']
  xpub6CataWdiZi0mdUeTPd8lT5or8nmkNcuyvz7WysvFKk84RwCD3YxvPEbvqAQY3rAPshwCmLoP2FMFMKHPJ4zeZXYVUHLv1VMrjPC7PW6w/<0;1/*,[73c5da0a/84/'0/'1']
  xpub6CataWdiZi0mdYfWtLxEQsAg1H9ooS1bmsJWuNB83FE1Fyk386WFycJgEzJqz3KsA5hd5oL2PubbMGCxFzC7hr16S72yqkG3HwZscyn/<0;1/*,[73c5da0a/84/'0/'2']
  xpub6CataWdiZi0mdBgNqGxj1G165X0tUax3PwJNi9WRjYgYf7VxmcfwXosR3v3oZLqR0bJ7dJbqJnCHtcfkaWmL9BUsW98y8KKJtChgW9y6n/<0;1/*;1))),or_i(and_v(v:after
    (1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0flac62dbd829b9a08ad),multi(2,[73c5da0a/84/'0/'3']
    xpub6CataWdiZi0mdHwRokjbyCrxddTDJgTyEaA0fKkbwU179LAWG5hJMCQ07BeJ4cmkbuXzdf45ZJK1qc1r3Z6EwEUDrUQbVdPkEc/<0;1/*,[b8688df1/84/'0/'0']
    xpub6DNfJehqF1LlUs9kwaQz12AJz9p9sVYtbGhtYqK01CYddkqV2vAYd0iWXSTdUjWhZyV1atV6NGKqGwgbLWq4a4XMRZ4mdpRjQDGS5ETV/<0;1/*,[b8688df1/84/'0/'1']
    xpub6DNfJehqF1LlUs0h1a4XLG3yuF10cRJ2k6B8ba97xvW6SwwY71NC6DLvkqfDLWuyCivJ4eMDxgftiXnazgp8rYvZQSKR3L8EGRs9asBarqce/<0;1/*;1))),or_i(and_v(v:older
      (65535),multi(2,[b8688df1/84/'0/'1']
      xpub6DNfJehqF1LlUwZ1dRfCQn365CDMPLSrqpSwaz4eRrtYz9qWdsVH9JqspH06yNGdG4VXGkbLgycKkXtWtB584iCuHcmj0mqR8z6NNXnpU51/<0;1/*,[b8688df1/84/'0/'3']
      xpub6DNfJehqF1LlUw8gR8sbt0R9LcAxwMdUw4678z5FZ7BLftwzQcT3yZb5e4w8UwA5JwbK3C2e9jykvXzggjPMBRKkC7z7b74uMv7Qp3TU/<0;1/*;1))),and_v(v:older(4255898)
        ,multi(1,[28645006/84/'0/'0']xpub6DBbzvudcQg2nS4tHskrm7F6kL64RwEY0j1GwBawmdNR0cJkH8KQdjuZ7TzDveib6B83deG6dCR6RjSopU8p1xv4awWQ0tIGbycAm
        /<0;1/*,[28645006/84/'0/'1']xpub6DBbzvudcQg2sPDRWpqtEn5VzPiwr1zNes4aHmMuog9Jmc2fc2J5fM2E39Y2y2iakmpWRpa3rXhZGnm4GKiJEXzmTctfQwBgD9ihVks
        /<0;1/*;1],[28645006/84/'0/'2']xpub6DBbzvudcQg2uk9RbxrscuXWjSysbfPYkPahQfmTfABfJ4j8TRxUnRd5SeGExfgPj63xcuTV9uny7p0BFbu3XKcN9rNxcNGRaDt9BD1gkBGZ
        /<0;1/*;1))))))#4ld0c8rxa
```

The watch-only card set:

```
# mnemonic bundle --descriptor-file policy.desc --network mainnet
# msl (omitted -- descriptor watch-only mode)

# mkl[0] (cosigner 0 xpub + origin)
mklqp tfcrz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkp eutks 2qvzq s7hd8 pnnmc v56c4 u
mklqp tfcrz pkg98 auetm d998g 9tyxu ae9vx n38r9 gtrp9 8q8s8 08l6s zjxkj t6r83 rk2jg 0cqns 0f30m txzd6 5mvwc ur9us v8qhc mu69g ql0zr c
mklqp tfcrz z74hw qxqdp 083je hp5td rfa0n 5zdfh w6425 sqe8p gsyxf tjtf4

# mkl[1] (cosigner 1 xpub + origin)
mklqp tfczz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkp eutks 2lcpa g7x7w 00uvz 59ukc 0
mklqp tfczz pszqg qzyqs zqqgz ypszq gqzqy 3zepu lhn90 du6se tz7wc ggw0m nt0ja rage0 ptr0v xfc3c 895gh kha0g k58rc wwcdc lrvxv vprnt a
mklqp tfczz zdp8b 2uxnp rxcue jaent sxshg r9vud frx5c c7npa egptf 64nsu d0nnn t334x du0p8 3dn95 g54tz

# mkl[2] (cosigner 2 xpub + origin)
mklqp tfcpz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkp eutks 2lcpa g07c7 z8k5q ayd8z 5
mklqp tfcpz pszqg qzyqs zqqgz yszqz gqzqy 3zepu lhn90 d76en fqwmt ug4zd pn8r9 4nrhg nwycc 6l2er rqlch 68l94 cdsr0 vxvng vzwq0 fhe86 j
mklqp tfcpz zappq c34fl c5lnh z7dzk jmwzf htn2t f9tp8 385pw pa2rh 2p9a6 flzxm hl43h hx3cm gcgpj6 25l8l

# mkl[3] (cosigner 3 xpub + origin)
mklqp tfcqz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkp eutks 2lcpa gqk6f 7csue 6uzgn a
mklqp tfcqz pszqg qzyqs zqqgz yrszq gqzqy 3zepu lhn90 d6awk dsm10 jm7xj ylhyy p7esa klkwk kasq2 4gnq3 578mn e3sjr 64gda uj68u hy768 d
mklqp tfcqz zzqp0 dmvs1 cy68s h0lw9 3nwwx th488 eqjvn 67gqj px7fy f84xj 7zjfc 29xjt agpg8 sxz5x lu0rx

# mkl[4] (cosigner 4 xpub + origin)
mklqp tfc8z qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkz ux3r0 3qvzq snwq0 e2f7m mwltn x
mklqp tfc8z pkg9w 55t7u h3mhp gjm2c tzuev mwdff ap2f5 zl9lk dthf3 72tg8 y9k3d 35qm0 06q30 450k4 vmqva kpyvw 7pkda jnnrd 36ymd aqkns v
mklqp tfc8z zm36y 9wa86 gpapq 5yppp y22y3 ww23f xkwgj cskqj sga8s gz6tl

# mkl[5] (cosigner 5 xpub + origin)
mklqp tfcxz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkz ux3r0 3lcpa gnlxn hxxfp rlea7 4
mklqp tfcxz pszqg qzyqs zqqgz ypszq gqzqy 3zepa 6j30m jj26l 5afq2 2n2ct 2h3zw rjps6 fyfqr 5zrz5 70jdm tqvx7 wwt0z 6a3fm fdmqr 0khu2 3
mklqp tfcxz zdpq6 ttmc1 y57ac 6vels 0u3jj 86dd5 pe7m3 55rtt w8r3a 2s45t szcmr x9xa5 qyec4 g3cru t8d54

# mkl[6] (cosigner 6 xpub + origin)
mklqp tfc9z qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkz ux3r0 3lcpa gz84k 67v3r 27gvy w
mklqp tfc9z pszqg qzyqs zqqgz yszqz gqzqy 3zepa 6j30m jemst 4u6j4 atq6u hd0m3 m5l9l vu9cd e6pcz wdrmf 3ejs5 4lts9 qugnu fy3yg 6r2y0 f
mklqp tfc9z zvqpq dye4j 7rexg zc7vs ch5dx s9e58 7e8u2 2q38u jtrt5 4lm26 dq30k fqqvm 7wyvx mtzy3 8suff

# mkl[7] (cosigner 7 xpub + origin)
mklqp tfcyz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkz ux3r0 3lcpa gd0hp xp2e6 dx8r4 8
mklqp tfcyz pszqg qzyqs zqqgz yrszq gqzqy 3zepa 6j30m jstc3 wnact 9lauq xytd7 f9xhq mtm1m a4zhs vtx5p 6e3s3 rggpd ce7mx am45j 76f5q d
mklqp tfcyz z3qpz lzuvw w4vzv ckseq udf17 d8u34 vrr7t 6kyvj lagme p2ywk 6q2q5 h3z2t vy9m0 6y2d 6kfgs
```

```
# mkl[8] (cosigner 8 xpub + origin)
mklqp tfctz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkq 5xg5q xqvzq s9n3l 96x69 pda2y s
mklqp tfctz pkg0d qcwzp syrk9 a808d 5f0ge nnj3x 9vjd5 63rqg zndx7 8e533 rulqy zd58c mvq4n udnky 9jhqs h5zpe maskg 7pdgx cftuk rgv6n 5
mklqp tfctz zt4c4 w6rtn jmnht 889lt vhne3 ktx5e ncnny 6spl8 gsqf5 3tn98

# mkl[9] (cosigner 9 xpub + origin)
mklqp tfc2z qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkq 5xg5q xlcpa g9zzk txfdl eumyf r
mklqp tfc2z pszqg qzyqs zqgqz ypsqz gqzqy 3zepa 5rpcg x0jla g0vc8 5ztty 3las6 2hfw7 3f3j8 zx6t3 lmtep 8skfu jv0xf u3hsl fcdrr rp8me 0
mklqp tfc2z zlcpn 80c5z mmvnx qknng p3d27 9fuyw f7zfw 5kqe8 wv93v 9fvn0 m39u6 5adln vgwul lstsp xm44n

# mkl[10] (cosigner 10 xpub + origin)
mklqp tfcfz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkq 5xg5q xlcpa g56yx xwr4a sa24n c
mklqp tfcfz pszqg qzyqs zqgqz ypsqz gqzqy 3zepa 5rpcg x0jla g0vc8 5ztty 3las6 2hfw7 3f3j8 zx6t3 lmtep 8skfu jv0xf u3hsl fcdrr rp8me 0
mklqp tfcfz z2yps arcmr l7tgu tgc8u f4m7g fcd8l fv33k nprwt hl06a m5gdw 9aqnf x4jz8 l2kjr 0p0ur 43e8j

# md1 (multisig wallet policy)
mdlff umxts 9z3m6 jzqaa fpr80 2gfga afp9n h4yyp m6jzx w75sj 3m6jz t802g grh4y yvaaf p9gen j3xrx hltjx q
mdlff umxts gqpxx 2v6mq q85ys zv6a4 pxuus yh2un u8xqz 8naa2 r2akw ukvgm 42gws 7xkx9 k7ctq mfdfe 9crr4 q
mdlff umxts s5mng y26xz zqyn9 xddhp k7csp xdw6s nwwgz t4wf7 rnqpr e774p 4wm8w txyd6 4yxex 7jxul xvqvk c
mdlff umxts apudv vtdas 2de5z 9drq3 rg4jn xhqqq rll7x ppvax dwqq5 8sngv q9zdg eccpe utk5l pf37m lhphs r
mdlff umxt3 zpgtn chdq5 feutk szxu7 9mq9y hp5gm u2ms6 yd794 cdzxl z7ux3 r03s2 ry2qr f9qs7 84qr5 aw8s9 8
mdlff umxt3 wg5qx 52ry2 qrpdk szs22 ws2kg demj2 cd8zw j2skz x2wq0 qw70l 4q99v dyh5x 0zxwq 65s8c 69ev9 z
mdlff umxt3 4j5jr 7qyur 6vt76 esnw4 xmrk8 qe0yr 02mhq rqxsh ncevm s69k3 57he6 px5mr n2zz0 33uqk qk8lq l
mdlff umxt3 u4308 vvy88 ae4he w375v hs43h krycu znj6y tmt7h 5t2r3 u8dqp 82uxn prxcu ejudw s6tu9 ntlfd x
mdlff umxtj 8xdwq 6zaqv 4n34y v6nrr 6v8h9 q9d82 kwr34 7wwd9 mtdxy pee03 z5f5x vuvkk vwu4e plqpe dw7ye 5
mdlff umxtj vfhzv vd04v 33s0u tarlj 6uxcp hkqup qc34f lc5ln hz7dz kjmzw fhtn2 tf9tp 83xfs kawee p093t j
mdlff umxtj ks9c8 4gwaq yhhf8 ugmw8 t46ek r0a7t 0c6gn 7uss8 mxrkm a6e6m kqp24 zvxzn clwk7 4rja8 2tmur l
mdlff umxtj eucfc pazqp 0dmvs lcy68 sh0lw 93nww xth48 8eqjv n67gq jpx7f yf84x j7zjf c2wr6 04yur u2eq3 4
mdlff umxtn qams5 fd4v9 3vvkld hx557 s4y6p 0jlmx 45mcl 995rj zmgkc 6qdhh aqgh6 68m2k dsxy5 ru7p2 fk5qg s
mdlff umxtn fmvzg ueurv mm8w8 gs4m5 lfayq yzsss yy3fg j9ee2 9zj26 l5afq 22n2c t2h3z wrj46 trnch trjpt z
mdlff umxtn sxr9y 3yqws gv2re 7fhdv psmee edutt gqwj6 778e9 8hwxn x0url yv537 ntdqw 0kusp xl3wy yry2c 2
mdlff umxtn e55rt tw8r3 a2s45 tszcm rrfms t4u6j 4atq6 uhd0m 3m5l9 lvu9c de6pc swdrm 73cph 9wc0u tcf6l 5
mdlff umxt5 x2z2h awq5r zqgrf xdvhs 7fjqk 8nyx9 arf5p wdpk flzjs yflyj c6a90 76kng ytumw tunf5 hft8g c
mdlff umxt5 wtstc 3wnac t9lau qxytd 7f9xh qmtml ma4zh svtx5 p6e3s 3rggp 2ezqz 979cu ua2mf vauyk 5ut07 p
mdlff umxt5 nqn9x 5xg8r 20lnf lydtq clj74 3ynhl 29kg2 3r4kj zs99q yrk9a 808d5 f0gen njsls rzrnn r484e 6
mdlff umxt5 uc4jf kr2yv pqd25 mclxj xy0nu qsfks lrdss k03kw csk2u zz7sg 880kz er6aw 9tk9y pxcwz cyw6f y
mdlff umxt4 zrtnj mnht8 89ltv hne3k tx5e8 uhl2r mxpaq j6ey0 lvxj4 6th52 vv3c3 kju07 67gdj m2hna x3x5p w
mdlff umxt4 f8skf ujuv0x falsr xwl3g 9hkcc vpd8x srz64 u2ncg unuyj afvpj wuctz c2jex lhzkn tnnzz u3sml 7
mdlff umxt4 30x4t nf7n9 z8y7v 6ypn2 35vam 58em2 v583v n4e8s ye3c6 4t8u9 28ma9 zqcw3 udszu lgfsx u8xey l
mdlff umxt4 lluk3 chss0 cnthu sns60 7jerr dxzxo h07l4 mhgs6 ut6px jq3gs 642g7 8tg45

# === Wallet bundle: descriptor, mainnet ===
# Threshold: 3 of 11
# Cosigners:
# @0: (no ms1; watch-only),mkl:5a703 (73c5da0a @ m/84'/0'/0')
# @1: (no ms1; watch-only),mkl:5a702 (73c5da0a @ m/84'/0'/1')
# @2: (no ms1; watch-only),mkl:5a701 (73c5da0a @ m/84'/0'/2')
# @3: (no ms1; watch-only),mkl:5a700 (73c5da0a @ m/84'/0'/3')
# @4: (no ms1; watch-only),mkl:5a707 (b8688df1 @ m/84'/0'/0')
# @5: (no ms1; watch-only),mkl:5a706 (b8688df1 @ m/84'/0'/1')
# @6: (no ms1; watch-only),mkl:5a705 (b8688df1 @ m/84'/0'/2')
# @7: (no ms1; watch-only),mkl:5a704 (b8688df1 @ m/84'/0'/3')
# @8: (no ms1; watch-only),mkl:5a70b (28645006 @ m/84'/0'/0')
# @9: (no ms1; watch-only),mkl:5a70a (28645006 @ m/84'/0'/1')
# @10: (no ms1; watch-only),mkl:5a709 (28645006 @ m/84'/0'/2')
# Template: descriptor
# md1: 5a70
# Recovery: any 3 of 11 signing keys + md1 (template card).
note: stdout is watch-only – public keys only, cannot spend
```

5.4 Restore round-trip -- the card set reconstructs the same first address

This 11-key, 4-branch policy is about as complex as a BIP-388 wallet gets, but the md1 cards carry it faithfully: take the md1 chunks and `restore` to read back the descriptor and its first address. The reconstructed first address is **identical** to the canonical descriptor's (section 5.2) -- proof the card set round-trips this whole policy without loss, every `after/older/sha256 lock` and `multi(...)` threshold across all four `or_i` branches preserved.

Note the contrast with the section 6 appendix: that `depth->=2 taproot` tree is the one shape the **shipped** mnemonic refuses to restore (it needs the `experimental/taproot-depth-ge2` branch). This degrading-multisig policy is fully supported on the real shipped binary -- the round-trip below uses it, no experimental branch required.

```
$ mnemonic bundle --descriptor-file policy.desc --network mainnet --json | jq -r ".md1[]" > policy.md1
note: stdout is watch-only – public keys only, cannot spend
```

```
$ mnemonic restore --network mainnet $(sed "s/^/--md1 /" policy.md1)
miniscript policy restore (11 cosigners)
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: wsh(or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,[73c5da0a
  ↪ /84'/0'/0']/xpub661MyMwAqRbcFHMVVPiBTXdc2Caj7vZhNFHJSgE59Aue2yYkXSrz5q9GaQ4rRjJVhHZtSiHWSzgmS5beaaTHWVmhpGC7SMdqMXHRXZi8as/<0;1>/*,[73c5da0a
```

```

↳ /84'/0'/1']xpub661MyMwAQBcGaxoYcLaxHHXZqEgSR0mN2P5ung8MJ8MNE535mLuhq7zjnrMkyA5eX6ehicVbU1FFPU39LGxbY8PmLPQxVRQmPfa3Q7spa/<0;1>/*,[73c5da0a
↳ /84'/0'/2']xpub661MyMwAQBcGuXAHBK3oquZS1HJiz2fVZ2idNcK4GLGTJyGZkPK7fv1N6euv5GzY18JD3WBG3SoLat23TLAVjhQMxDVMAqymQNhg3RFT8/<0;1>/*)),or_i(and_v
↳ (v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(2,[73c5da0a/84'/0'/3']
↳ xpub661MyMwAQBcGHLcZcLjg25oG8wSyqSE5XNM9uMks6vrpH4pRDC8UmAynovThuKraidMeEKJ2FcqBw1eF76aeu1vrGtLJXUiJXr4r9N1TZQ/<0;1>/*,[b8688df1/84'/0'/0']
↳ xpub661MyMwAQBcGowNgeNcLS8Cgl2vnZybpJqtkbCmSQMdq2qzcDWq3CXKg7x5BqvcNCsANuW6n1soN7JFK2j3HfxV57Nm2RLKo2UzHgbs6/<0;1>/*,[b8688df1/84'/0'/1']
↳ xpub661MyMwAQBcEv3U8uuxavsQ48LNNYwcNge8rT7SaMS558KiEwXoP72TQ8ARYjczPTtVQz6CxcaBTEE3XchmYvSiHcVbC9h17CmyfG7sVq9/<0;1>/*)),or_i(and_v(v:older
↳ (65535),multi(2,[b8688df1/84'/0'/2']
↳ xpub661MyMwAQBcG7Xht9EwgNucA47Rmgg8Bn5bNmFdJMkoTHQDXirpogQHkVNRcwaY6KwGnUYMUNBFCNaRq4WnsqWw2VNUdW6ymHXfVpk4c3/<0;1>/*,[b8688df1/84'/0'/3']
↳ xpub661MyMwAQBcEbqBvNkLuDtudGA2PHAbtWUuHKe3CKjZCaLjxLGS68Sjpw8CnXsj8xPGXaV9ZWL3j9ktbed8y1aeNVK95HrkgHfHG8XM5Eh/<0;1>/*)),and_v(v:older(4255898)
↳ ,multi(1,[28645006/84'/0'/0']xpub661MyMwAQBcEdBofBaGbnse74WRuyEbXRSzmzq8jzthzutDnXTV2yNQpZgs3ubwuNp7yrSHnECOA5xHgnOEDH4HSGWqLTydi6nWVZCfXPK
↳ /<0;1>/*,[28645006/84'/0'/1']xpub661MyMwAQBcH2WNbmtz4pZ8wDtpxdYoe6E4r5o8pXedve17srmalLCEjM8WcpV67xsc36KpBntYUdqo5dpcFMzRfzSZSa4C5DRty4eDNF
↳ /<0;1>/*,[28645006/84'/0'/2']xpub661MyMwAQBcGqcAAnB9mhvQsdUx2fKasUoXT2Gmpt2tFz94wRfAkhLhZUJkjQ5pgnd9Ny9EwrgCHbAASVnQShCbfnhGSAkAk2k6yGoWXAv
↳ /<0;1>/*))))))#jgule7j
first recv: bclq4g7564xxd9hj68hquw5e558cafshsklerkr0asfzqp6puq74veersp6qs5
cosigner @0: 73c5da0a [84'/0'/0'] from md1 (not independently verified)
cosigner @1: 73c5da0a [84'/0'/1'] from md1 (not independently verified)
cosigner @2: 73c5da0a [84'/0'/2'] from md1 (not independently verified)
cosigner @3: 73c5da0a [84'/0'/3'] from md1 (not independently verified)
cosigner @4: b8688df1 [84'/0'/0'] from md1 (not independently verified)
cosigner @5: b8688df1 [84'/0'/1'] from md1 (not independently verified)
cosigner @6: b8688df1 [84'/0'/2'] from md1 (not independently verified)
cosigner @7: b8688df1 [84'/0'/3'] from md1 (not independently verified)
cosigner @8: 28645006 [84'/0'/0'] from md1 (not independently verified)
cosigner @9: 28645006 [84'/0'/1'] from md1 (not independently verified)
cosigner @10: 28645006 [84'/0'/2'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only - public keys only, cannot spend

```

(As in section 6.3, restore re-serialises each key as a depth-0 xpub661My... -- a different descriptor string, identical addresses. Compare the `first recv:` line above with the BSMS first address in 5.2: byte-for-byte the same.)

5.5 Watch-only export for Nunchuk / Core / Sparrow

There is no dedicated `nunchuk` emitter, but Nunchuk imports miniscript wallets from a **descriptor** or a **BSMS (BIP-129)** record -- both shown above in 5.2 (--format descriptor for *Add Wallet* -> *Import* -> *descriptor*, and --format bsms for the multisig import format, which also Bitcoin Core and Sparrow accept). After import, fund the address; each spend path opens only when its lock matures, and the hashlock tiers additionally require revealing the secret word's preimage `x`.

6. Taproot version of the degrading wallet (the pathological example)

wsh(...) reveals the whole policy on every spend. **Taproot** gives a cooperative **key-path** spend (cheap, private, looks like single-sig) and splits the fallbacks into **script-tree leaves**, so a spend reveals only the leaf it uses. We keep the same four timelock/hash/multisig tiers as fallbacks and add a distinct cooperative internal key `kint` ([.../84'/0'/4'1]). Taproot multisig uses `multi_a`, not `multi`. **12 distinct keys** in total.

6.1 Why depth-1 (the one-tier-per-leaf limit on master)

The tidiest layout is one tier per leaf (4 leaves), but that is a **depth-2** taptree, and the shipped rust-miniscript pin mis-formats depth->=2 taptrees (the PR-#953 bug). The toolkit refuses such a descriptor up front rather than emit a malformed one. The four-leaf (depth-2) descriptor file:

```
$ cat taproot-4leaf.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNTnqr2UHCBEsCoL3eFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{and_v(v:
  ↳ after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↳ xpub6CatWdiZiodmUeTdP8LT5or8nmbKNcuyvz7WYksVfKKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLOp2fMFMKHPJ4ZeZXYVUhLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↳ xpub6CatWdiZiodmYVtWLTQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZS3KA5dh5Lo2PbubMGXcfZtRthV6ST2qquL9w3HSzcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↳ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNNi9WriJYgYf7VxcmuFxsRw3focZLGRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgWeryy6/<0;1>/*)),and_v(v:after
  ↳ (1893456000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↳ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsyEaaQfKjKwbWUi79LAWG5GhJMCQB7BeJc47MkubXuzdf45JZHk1qcr1GZ5EwREUDVDLVdpKec/<0;1>/*,[b8688df1/84'/0'/0']
  ↳ xpub6DNfJehqFILLUs9kwaqDu12Ajpz9psYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDujWHZVy1AtV6ENGKwGwBLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↳ xpub6DNfJehqFILLUsh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvKqfDLWuyCivJ4eMDXgFtiTxnazgp8rYvZQSKR3L8EGRS9asBarqpe/<0;1>/*)),{and_v(v:older
  ↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↳ xpub6DNfJehqFILLUwZ1dRfCQ365CDMPPLSsrrqpSwaz4eRrtYz9qWdsVH9JqshpQ6yNgdG4VXGKbLgckKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↳ xpub6DNfJehqFILLUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzwQCT3yZb5eW4U8AVJwbKcZ2iegjvJvKx2ggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*)),and_v(v:older(4255898)
  ↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvudcQg2nS4tH5krM7FGKXl6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuzTZJtwDveibGd83eGzdGCR6NjSqpu8p2Xx4wWQK7iGBycAm
  ↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvudcQg2sPDRWpqfEn5VzPiwrd1zNes4aHmnnUog9Jmc2fc2J5fM2E39Yz2iaKmpWRpa3rXhZGNmd4GKiJEZxmTcftQpWBGd9ihVks
  ↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABfJ4j8TRXUnRd5eGgEXfgPj63xcuTV9uny7pQBFbu3XKcN9rNxcNGRaDT9BD1gkBGZ
  ↳ /<0;1>/*)))})
```

Asking the toolkit to export it:

```
$ mnemonic export-wallet --descriptor "$(cat taproot-4leaf.desc)" --format descriptor --network mainnet
error: export-wallet script-type derive: taptree branch must have 2 children, but found 1
```

So we use a **depth-1** tree (2 leaves) and pack two tiers per leaf with `or_i`: Leaf A = tier 1 or tier 2 (the absolute-timelock + secret-word tiers); Leaf B = tier 3 or tier 4 (the relative-timelock tiers). (A rust-miniscript release > 13.1.0 containing #953 reopens deep trees -- tracked in FOLLOWUP taproot-coverage-cycle-on-miniscript-gt-13-1-0.)

6.2 Build + validate

The hand-written depth-1 `tr(...)` descriptor file:

```
$ cat taproot.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNTnqr2UHCBEsCoL3eFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{or_i(
  ↳ and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↳ xpub6CatWdiZiodmUeTdP8LT5or8nmbKNcuyvz7WYksVfKKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLOp2fMFMKHPJ4ZeZXYVUhLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↳ xpub6CatWdiZiodmYVtWLTQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZS3KA5dh5Lo2PbubMGXcfZtRthV6ST2qquL9w3HSzcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↳ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNNi9WriJYgYf7VxcmuFxsRw3focZLGRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgWeryy6/<0;1>/*)),and_v(v:after
  ↳ (1893456000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↳ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsyEaaQfKjKwbWUi79LAWG5GhJMCQB7BeJc47MkubXuzdf45JZHk1qcr1GZ5EwREUDVDLVdpKec/<0;1>/*,[b8688df1/84'/0'/0']
  ↳ xpub6DNfJehqFILLUs9kwaqDu12Ajpz9psYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDujWHZVy1AtV6ENGKwGwBLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↳ xpub6DNfJehqFILLUsh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvKqfDLWuyCivJ4eMDXgFtiTxnazgp8rYvZQSKR3L8EGRS9asBarqpe/<0;1>/*))),or_i(and_v(v:older
  ↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↳ xpub6DNfJehqFILLUwZ1dRfCQ365CDMPPLSsrrqpSwaz4eRrtYz9qWdsVH9JqshpQ6yNgdG4VXGKbLgckKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↳ xpub6DNfJehqFILLUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzwQCT3yZb5eW4U8AVJwbKcZ2iegjvJvKx2ggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*)),and_v(v:older(4255898)
  ↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvudcQg2nS4tH5krM7FGKXl6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuzTZJtwDveibGd83eGzdGCR6NjSqpu8p2Xx4wWQK7iGBycAm
  ↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvudcQg2sPDRWpqfEn5VzPiwrd1zNes4aHmnnUog9Jmc2fc2J5fM2E39Yz2iaKmpWRpa3rXhZGNmd4GKiJEZxmTcftQpWBGd9ihVks
  ↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABfJ4j8TRXUnRd5eGgEXfgPj63xcuTV9uny7pQBFbu3XKcN9rNxcNGRaDT9BD1gkBGZ
  ↳ /<0;1>/*)))})
```

Validate and canonicalise it. The full canonical descriptor, every xpub in full:

```
$ mnemonic export-wallet --descriptor "$(cat taproot.desc)" --format descriptor --network mainnet
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNTnqr2UHCBEsCoL3eFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{or_i(
  ↳ and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↳ xpub6CatWdiZiodmUeTdP8LT5or8nmbKNcuyvz7WYksVfKKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLOp2fMFMKHPJ4ZeZXYVUhLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↳ xpub6CatWdiZiodmYVtWLTQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZS3KA5dh5Lo2PbubMGXcfZtRthV6ST2qquL9w3HSzcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↳ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNNi9WriJYgYf7VxcmuFxsRw3focZLGRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgWeryy6/<0;1>/*)),and_v(v:after
  ↳ (1893456000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↳ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsyEaaQfKjKwbWUi79LAWG5GhJMCQB7BeJc47MkubXuzdf45JZHk1qcr1GZ5EwREUDVDLVdpKec/<0;1>/*,[b8688df1/84'/0'/0']
  ↳ xpub6DNfJehqFILLUs9kwaqDu12Ajpz9psYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDujWHZVy1AtV6ENGKwGwBLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↳ xpub6DNfJehqFILLUsh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvKqfDLWuyCivJ4eMDXgFtiTxnazgp8rYvZQSKR3L8EGRS9asBarqpe/<0;1>/*))),or_i(and_v(v:older
  ↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↳ xpub6DNfJehqFILLUwZ1dRfCQ365CDMPPLSsrrqpSwaz4eRrtYz9qWdsVH9JqshpQ6yNgdG4VXGKbLgckKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↳ xpub6DNfJehqFILLUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzwQCT3yZb5eW4U8AVJwbKcZ2iegjvJvKx2ggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*)),and_v(v:older(4255898)
  ↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvudcQg2nS4tH5krM7FGKXl6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuzTZJtwDveibGd83eGzdGCR6NjSqpu8p2Xx4wWQK7iGBycAm
  ↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvudcQg2sPDRWpqfEn5VzPiwrd1zNes4aHmnnUog9Jmc2fc2J5fM2E39Yz2iaKmpWRpa3rXhZGNmd4GKiJEZxmTcftQpWBGd9ihVks
  ↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABfJ4j8TRXUnRd5eGgEXfgPj63xcuTV9uny7pQBFbu3XKcN9rNxcNGRaDT9BD1gkBGZ
  ↳ /<0;1>/*)))})
```

```

↳ ([73c5da0a/84'/0'/4']
↳ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqShpQ6yNGdG4VXGKbLgCkKkXtWtB584iCuHcmjomqR8z6NNxNpU51/<0;1>/*, [b8688df1/84'/0'/3']
↳ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzQcT3yZb5eW4U8AVJwbKcZ2iegjvjKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*), and_v(v:older(4255898)
↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvdudcQg2nS4tH5Krm7FGkXL6arWxEYoZ1g1GwBaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpU8p2Xx4wMQK7iGBycAm
↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvdudcQg2sPDRWpqfEn5VzPiwrd1zNes4aHmNUog9Jmc2fc2J5fM2E39Y2yZiakmpWRpa3rXhzGNmd4GKiJEXZmTcftQpwBgD9ihVks
↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvdudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABf4j8TRXUnRd5eGgEXfgPJ63xcuTV9uny7pQBfBu3XKcN9rNxcNGRaDT9BD1gkBGZ
↳ /<0;1>/*)))#nerswx7
note: stdout is watch-only – public keys only, cannot spend

```

`mint ([73c5da0a/84'/0'/4'])` is the key-path; the two `or_i(...)` blocks are the two script leaves; `after(...)` are the absolute (height/time) locks and `older(...)` the relative (blocks/time) locks -- the same four kinds as section 5.

6.3 Engrave + first address

Every key is distinct, so it engraves (watch-only). Take the `md1` chunks and restore to read the first address (this round-trip also proves the **real internal key** at the trunk reconstructs -- the non-NUMS internal-key feature, shipped in `v0.55.3`). The descriptor file (shown again):

```

$ cat taproot.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmXswr13Gd5aNtNqr2UHCBEsCoL3eEFVaM7n8kY5K4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,[or_i(
↳ and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
↳ xpub6CatWdiZiodmUeTdp8LT5or8nmbKNcuYv7WYksVfKKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLOp2fMFMKHPJ4ZeXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
↳ xpub6CatWdiZiodmYVtWLTtEQAsAg1H9ooS1bmsJUBWQ83FE1Fyk386FWcyicJgEzV3quZSJKASdh5Lo2PbubMgXcfZtRthV6S2tqqL9w3H5szcUn/<0;1>/*,[73c5da0a/84'/0'/2']
↳ xpub6CatWdiZiodmBgQcQxxjGN165QxTU4PwNNi9WrjYgYf7VxcmuFxsRw3focZLgRbDjDjJbqZhPCTfkcaWmL9BuSw98ybKJktcHgweryy6/<0;1>/*)),and_v(v:after
↳ (1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
↳ xpub6CatWdiZiodmHurRokjbyCcxrdTDJgTsyEaaQfKjKwbWUi79LAWG5GhJMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLvdPkEC/<0;1>/*,[b8688df1/84'/0'/0']
↳ xpub6DNfJehqF1LU59kwaqDu12Ajpz9psYVtbGhTyQk0lCYdkkV2yAr4RD1wXSTTDjUjWhzVy1AtV6ENGKwgbLWqa4wXMR4ZmdpRjQ8G5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
↳ xpub6DNfJehqF1LUsoh1a4XLG3yuFY1oCRJd68Kba97xvW6SwvY71Nc6LvqfDLWuyCivJ4eMDXgft1Txnazgp8rYvZSQKR3L8EGrs9asBarqpe/<0;1>/*))),or_i(and_v(v:older
↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
↳ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqShpQ6yNGdG4VXGKbLgCkKkXtWtB584iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
↳ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzQcT3yZb5eW4U8AVJwbKcZ2iegjvjKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*),and_v(v:older(4255898)
↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvdudcQg2nS4tH5Krm7FGkXL6arWxEYoZ1g1GwBaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpU8p2Xx4wMQK7iGBycAm
↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvdudcQg2sPDRWpqfEn5VzPiwrd1zNes4aHmNUog9Jmc2fc2J5fM2E39Y2yZiakmpWRpa3rXhzGNmd4GKiJEXZmTcftQpwBgD9ihVks
↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvdudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABf4j8TRXUnRd5eGgEXfgPJ63xcuTV9uny7pQBfBu3XKcN9rNxcNGRaDT9BD1gkBGZ
↳ /<0;1>/*)))

```

```

$ mnemonic bundle --descriptor-file taproot.desc --network mainnet --json | jq -r ".md1[]" > taproot.md1
note: stdout is watch-only – public keys only, cannot spend

```

```

$ cat taproot.md1
md1f40gpvqz6jzwrh4yypm6jzwx75j3m6jzt802ggrh4yyvaafp9rh4yykw75s802gemcn9wrf4m7m5yt9
md1f40gpvqzj2z2sqrqgy2efntvqq7sjqfntk5ymnjqatj0suscqg70h4gdtkemj3r4f6p6rc6cnxqhq4hnx7rv
md1f40gpvqscmmq5mngy26gz2zg6v6mwrda3qz6a4pxuusyhtunu8xqz8naa2r2akwukvgm4g7w7kdyaf805mf
md1f40gpvq6gws7xx9k7c9xu6pzkjgq9j9efntsqpplllyqshsnxhqgrcf5gqzn2cvasqu79cwukltfwjcnckn
md1f40gpvpgr9p0w2a5z3883w6pgmncdq53eutks2twrg3hckhp5gmuts6y7d9xcdzlxry5xgklamn3a93x43x
md1f40gpvpg5q5x52ryzq7rt9p9qpskfufq44hccjz40ltnk8tdswlxxk6gs50wsf6dvu4jev3eqgy8f0nxdna6x
md1f40gpvp56gl4vs3t2qswfth67sept4g4hf5939sedkqhey4nnc0chj6t2x82xgqygmpkv7cuw0m7akf723
md1f40gpvp622ws2kgdemj2cd8zwj2skzx2wq0q70l4q99vdyh5x08v4ysls8p9nzlkkvym5v1rcugrdkju72
md1f40gpvz9fka3cxtetgm6kacpcp59u7xtux3d5d847ws4xev6setz7wcgwg0mt0jarageqgfesknkmcceqj
md1f40gpvz0ptr0vx3c9895ghka0gk58rcw6qzw4cdxzd3en9mnhxqdpws2ec6jxd33axqmwg8tzlv563mv
md1f40gpvznmjszkn4t8pc6l88xna4nqjuhuhc32y6rxwtt8w3uxf33474jxxpl30507ttsmq84l460gwc0g4w
md1f40gpvzcx7rcsyrz48lzn7wute526tvfexawdfdy4vy7y7s9c84gwagyhfh8ugmwf746ekqd54ka5c8c48mx
md1f40gpvr0a7t0c6gn7uss8mxrkma6e6mkap24zvzxnc1w0xxzg0gsqtmwmy87px3u9mlm3vgd8s0y6hrrvud
md1f40gpvrvmnj3aaf7gyny7hjyqsfhjzfaf5hs5jwzh3mhpgjm2ct2uevmwdfap2f5z19lsqqkg20r8nj9q
md1f40gpvrkdtfh37zr2qnhxzcK5fhfac7j2Lxnax2ywfue5grx4rgemgh0nk5eg0ze8tj0qfnr3syntujt2jg5xpl
md1f40gpvr6zg55fzuu4z339d06w5s99f4v94tc38peqc5jysp6pp3g08exa4srx0889h3ddqqnqhd12vm2eucf
md1f40gpvpv6ttmcl1y57ac6vels0u3jj86dd5pe7m355rttw8r3a2s45tsczmremst4u6j4atqccq4mtqrtm0ny
md1f40gpvyg6uhd0m3m5l9lvu9cde6pcswdr73ejs4lts9qcszq6fnt9u8jvs93uep30g6dqg0k3hj4xq9v7xn
md1f40gpvynng0aj0c55p20eykxhftlk456pzlvcqh3za8mskltmcqvgkmu72dwpkhhlhm290qceu48zwu553kp
md1f40gpvyckdgr4nrpzsxs4jyqut3ee64sfnz6ryr348le5ljx4sv0e02c2jfm14zmy9g36c6spmkzfw55f
md1f40gpv9rfpgzjjzpmz7nhnk6yh5veeegnzkfxcgdg3sypfkno8u6gc370szpx6rudkq2e7xecrr560mc7uy987
md1f40gpv9tzzetsg6tpqua7cty0t4c4w6rtijnmht8891lvhne3ktx5etuh12rmxpaqj6ey0lg6d4mp519hjrne
md1f40gpv95xj46th52vv3c3kju0767gfu9j0ynrejo0uqenhu2pdakxrrtfe5qck40z57z8ylqssej5h2v929gq
md1f40gpv9eyh2tqvnhxzcK5fhfac7j2Lxnax2ywfue5grx4rgemgh0nk5eg0ze8tj0qfnr3syntujt2jg5xpl
md1f40gpvx92k0c250h62ypsarcmr17tgutcg8uf4m7gfc8l1fv33knpwrthl06am5gdw9a9nqfkhq0hnyzweea

```

```

$ mnemonic restore --network mainnet $(sed "s/~/-md1/" taproot.md1)
miniscript policy restore (12 cosigners)
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: tr([73c5da0a/84'/0'/4']xpub661MyMwAQrbCYUkSqsBgaz1Lob8pCa1rM1SJBCEGcYyP9LisxZ2m1GoqDj137XvHdY2NkctqiElixaAFqYHf91CFpFpKicVb7TzvrGSe
↳ /<0;1>/*,[or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a
↳ /84'/0'/0']xpub661MyMwAQrbCFHwVpCiBTxd2Caj7vZnHFHJ5G59Aue2YkXsRz5q9Ga0rRjJvHHTZtCiHwS2gm55beaaTHwMhpGc7SMdqMHRXZi8as/<0;1>/*,[73c5da0a
↳ /84'/0'/1']xpub661MyMwAQrbC6axoYcLaxHHXZgEqSRQmN2P5ung8M38MNE535mLuq7zjnrmKYA5eX6ehicVbU1FFPU39LGxbY8PmPLQxVRQmPfa3Q7spa/<0;1>/*,[73c5da0a
↳ /84'/0'/2']xpub661MyMwAQrbC6uXAHBK3oquZS1HJiz2fVZ2idNC4G6LGTJyGZgZPK7fv1N6euv5GzY18JD3WBG3SoLat23TLAVjhQMXdVMAqymNHg3RFT8/<0;1>/*))),and_v(v:
↳ after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
↳ xpub661MyMwAQrbCGLCZcljg25oG8w5yqS5EXNM9uMks6vrpH4pRDC8UAmaynovThuKraidMeEKJ2fCqbw1eF76aeu1vrGtLXUjIXr4r9N1TQZ/<0;1>/*,[b8688df1/84'/0'/0']
↳ xpub661MyMwAQrbC6owNgeNcL58Cgl2vnyZybpJqtkbCmSQMdqZqcDwqq3CXGx7x5BqvcNC5ANuW6n1soN7JFK2j3HfxV57Nm2RLKo2UzHgbs6/<0;1>/*,[b8688df1/84'/0'/1']

```

```

    ↪ xpub661MyMwAQbRcEv3U8uuxavsQa8LNNYwcNge8rT7SaMS558KiEwXoP72TQ8ARYjczPTtVQz6CxcaBTEE3XchmYvSiHcVbC9h17CmyfG7sVq9/<0;1>/*))) ,or_i( and_v(v:older
    ↪ (65535), multi_a(2, [b8688df1/84'/0'/2']
    ↪ xpub661MyMwAQbRcG7Xht9EwgNucA47Rmgg8Bn5bNmFdJmKotHQDXirpogQHkVNRcAy6KwGnUYUMNBFcNaRq4WnsqWw2VNUddW6ymHXfVpk4c3/<0;1>/*, [b8688df1/84'/0'/3']
    ↪ xpub661MyMwAQbRcEbqBNKLdUtudGA2PHAbtWuUHKe3CKjZCaLjLGS68SjpwBCnXsj8xPGXaV9ZWL3j9ktbed8y1aeNVK95HrkGfHfGBXMS5Eh/<0;1>/*))) ,and_v(v:older(4255898)
    ↪ , multi_a(1, [28645006/84'/0'/0'] xpub661MyMwAQbRcEdBoFbaBgbnsg74WRuyEbXRsSmzq8jzthztDnXTV2yNQPzgs3ubwuNp7yrSHnECa5xHgnEDH4HSGWqLtYdi6nWZCfXPk
    ↪ /<0;1>/*, [28645006/84'/0'/1'] xpub661MyMwAQbRcH2WNMBt24pZ8WdtpxdYoe6E4r5o8rXedve17srmalLCEjM8WcpV6k7xsc36KpBNtYUdqo5dpCFmZrZf5S2a4C5DRty4eDNF
    ↪ /<0;1>/*, [28645006/84'/0'/2'] xpub661MyMwAQbRcGqcAAnB9mhvQsdUx2fKasUoXT2Gmpt2tFz94wRfAkhuLhZUJkjQ5pgnd9Ny9EwrgcHbAASVnQShCbfnhGskAk2k6yGowXAv
    ↪ /<0;1>/*)))#7cy3x3q9
first recv: bc1p9stcwz5597fmkxae9343k8edzkcvdcz9qp65r6p447pg0et82yqst3d2c
cosigner @0: 73c5da0a [84'/0'/4'] from md1 (not independently verified)
cosigner @1: 73c5da0a [84'/0'/0'] from md1 (not independently verified)
cosigner @2: 73c5da0a [84'/0'/1'] from md1 (not independently verified)
cosigner @3: 73c5da0a [84'/0'/2'] from md1 (not independently verified)
cosigner @4: 73c5da0a [84'/0'/3'] from md1 (not independently verified)
cosigner @5: b8688df1 [84'/0'/0'] from md1 (not independently verified)
cosigner @6: b8688df1 [84'/0'/1'] from md1 (not independently verified)
cosigner @7: b8688df1 [84'/0'/2'] from md1 (not independently verified)
cosigner @8: b8688df1 [84'/0'/3'] from md1 (not independently verified)
cosigner @9: 28645006 [84'/0'/0'] from md1 (not independently verified)
cosigner @10: 28645006 [84'/0'/1'] from md1 (not independently verified)
cosigner @11: 28645006 [84'/0'/2'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only – public keys only, cannot spend

```

(Restore re-serialises each key as a depth-0 xpub661My... -- a different descriptor string, identical addresses; that is how the md1 card stores keys.)

6.4 Export for wallets (Nunchuk / Core / Sparrow)

descriptor and bitcoin-core both work for taproot. The descriptor file (shown again):

```

$ cat taproot.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmXswr13Gd5aNtNqr2UHCBEsCoL3eEFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XmGcQNeQ4HcjfVTn9TuXe/<0;1>/*, {or_i(
    ↪ and_v(v:after(1000000), and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad), multi_a(3, [73c5da0a/84'/0'/0']
    ↪ xpub6CatWdiZiodmUeTdp8LT5or8nmbKNcuYvz7WYksVFkKB4RHwCD3XyuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*, [73c5da0a/84'/0'/1']
    ↪ xpub6CatWdiZiodmYVtWltEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZ5JKA5dh5Lo2PbubMGxCFzTtRthV6ST2qquL9w3H5zcUn/<0;1>/*, [73c5da0a/84'/0'/2']
    ↪ xpub6CatWdiZiodmNBgqcQxxjGN165QxTU4PwNNi9WrijYgYf7VxcmuFxoSrw3focZLgRdbjDjJbqZhPCTfkcaWmL9BuSw98ybKKJtcHgwWeryy6/<0;1>/*))) ,and_v(v:after
    ↪ (1893456000), and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad), multi_a(2, [73c5da0a/84'/0'/3']
    ↪ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsYEAaQfKjKwBwUi79LAWG5gHjMCQB7BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLvdPkEC/<0;1>/*, [b8688df1/84'/0'/0']
    ↪ xpub6DNfJehqF1LU59kwaQDu12Ajz9p9sYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDjWhZVy1AtV6ENGKwgbLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*, [b8688df1/84'/0'/1']
    ↪ xpub6DNfJehqF1LUsoh1a4XLG3yuF1oCRJd6Bkba97xvW6SwvY71Nc6LvKqFDLWuyCivJ4eMDxgFtiTxnazgp8rYvZQSKR3L8EGRs9asBarqpe/<0;1>/*))) ,or_i( and_v(v:older
    ↪ (65535), multi_a(2, [b8688df1/84'/0'/2']
    ↪ xpub6DNfJehqF1LUwZ1drFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqShpQ6yNgdG4VXGKbLgxcKkXtWtB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*, [b8688df1/84'/0'/3']
    ↪ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzWQct3yZb5eW4U8AVJwbKcZ2iegjvkvX2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*))) ,and_v(v:older(4255898)
    ↪ , multi_a(1, [28645006/84'/0'/0'] xpub6DBbzvudcQg2nS4tH5Krm7FGkXL6arWxEyoZ1g1GwbaNdwRobcJmxH8KQdjuzTJ2twDveibGd83eGzdGCR6NjSqpU8p2Xx4wWQK7iGBycAm
    ↪ /<0;1>/*, [28645006/84'/0'/1'] xpub6DBbzvudcQg2sPDRWpqfEn5VzPiwrldzNes4aHmNUog9Jmc2fc2J5fM2E39YzY2iakmpWRpa3rXhzGNmd4GKiJEZxmTctfQpwBGd9ihVks
    ↪ /<0;1>/*, [28645006/84'/0'/2'] xpub6DBbzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABFJ4j8TRXUnRd5eGgEXfgPJ63xcuTVun9y7pQBfBu3XKCN9rNxcNGRaDT9BD1gkBGZ
    ↪ /<0;1>/*)))#7cy3x3q9

```

Bitcoin Core `importdescriptors` payload:

```

$ mnemonic export-wallet --descriptor "$(cat taproot.desc)" --format bitcoin-core --network mainnet
{
  "active": true,
  "desc": "tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmXswr13Gd5aNtNqr2UHCBEsCoL3eEFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XmGcQNeQ4HcjfVTn9TuXe/<0;1>/*,{
    or_i( and_v(v:after(1000000), and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad), multi_a(3, [73c5da0a/84'/0'/0']
    ↪ xpub6CatWdiZiodmUeTdp8LT5or8nmbKNcuYvz7WYksVFkKB4RHwCD3XyuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*, [73c5da0a/84'/0'/1']
    ↪ xpub6CatWdiZiodmYVtWltEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZ5JKA5dh5Lo2PbubMGxCFzTtRthV6ST2qquL9w3H5zcUn/<0;1>/*, [73c5da0a/84'/0'/2']
    ↪ xpub6CatWdiZiodmNBgqcQxxjGN165QxTU4PwNNi9WrijYgYf7VxcmuFxoSrw3focZLgRdbjDjJbqZhPCTfkcaWmL9BuSw98ybKKJtcHgwWeryy6/<0;1>/*))) ,and_v(v:after(1893456000)
    ↪ , and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad), multi_a(2, [73c5da0a/84'/0'/3']
    ↪ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsYEAaQfKjKwBwUi79LAWG5gHjMCQB7BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLvdPkEC/<0;1>/*, [b8688df1/84'/0'/0']
    ↪ xpub6DNfJehqF1LU59kwaQDu12Ajz9p9sYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDjWhZVy1AtV6ENGKwgbLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*, [b8688df1/84'/0'/1']
    ↪ xpub6DNfJehqF1LUsoh1a4XLG3yuF1oCRJd6Bkba97xvW6SwvY71Nc6LvKqFDLWuyCivJ4eMDxgFtiTxnazgp8rYvZQSKR3L8EGRs9asBarqpe/<0;1>/*))) ,or_i( and_v(v:older
    ↪ (65535), multi_a(2, [b8688df1/84'/0'/2']
    ↪ xpub6DNfJehqF1LUwZ1drFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqShpQ6yNgdG4VXGKbLgxcKkXtWtB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*, [b8688df1/84'/0'/3']
    ↪ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzWQct3yZb5eW4U8AVJwbKcZ2iegjvkvX2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*))) ,and_v(v:older(4255898) ,
    ↪ multi_a(1, [28645006/84'/0'/0'] xpub6DBbzvudcQg2nS4tH5Krm7FGkXL6arWxEyoZ1g1GwbaNdwRobcJmxH8KQdjuzTJ2twDveibGd83eGzdGCR6NjSqpU8p2Xx4wWQK7iGBycAm
    ↪ /<0;1>/*, [28645006/84'/0'/1'] xpub6DBbzvudcQg2sPDRWpqfEn5VzPiwrldzNes4aHmNUog9Jmc2fc2J5fM2E39YzY2iakmpWRpa3rXhzGNmd4GKiJEZxmTctfQpwBGd9ihVks
    ↪ /<0;1>/*, [28645006/84'/0'/2'] xpub6DBbzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABFJ4j8TRXUnRd5eGgEXfgPJ63xcuTVun9y7pQBfBu3XKCN9rNxcNGRaDT9BD1gkBGZ/<0;1>/*)))#q3s6uZuk",
    "internal": false,
    "range": [
      0,
      999
    ],
    "timestamp": 0
  },
  {
    "active": true,

```

```
"desc": "tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aMtNqr2UHCBEsCol3eEFVaM7n8kY5kS4daaP83gW0ncmzL3Wzt79mEiLiX6XZs6X0mGcQNeQ4HcjfVTn9TuXE/1/*,{
  ↪ or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↪ xpub6CatWdiZiodmUeTp8LT5or8nmbKNcuyvz7WYksVFkKB4RHwCD3XyuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/1/*,[73c5da0a/84'/0'/1']
  ↪ xpub6CatWdiZiodmYVtWltEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3qzSJKASdh5Lo2PbubMGxCfZtRthV6ST2qquL9w3HSzcUn/1/*,[73c5da0a/84'/0'/2']
  ↪ xpub6CatWdiZiodmNGqcQxxjGN1650xTU4PwNNi9WrijYgYf7VxcmuFosRw3focZLgRbdJdJbqZhPCTfkcaWmL9BuSw98ybKKJtcHgWeryy6/1/*)),and_v(v:after(1893456000)
  ↪ ,and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↪ xpub6CatWdiZiodmHurRokjbycCrxdTDJgTsyEaaQfKjKbWUi79LAWG5gHjMCQB7BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLvdPkEC/1/*,[b8688df1/84'/0'/0']
  ↪ xpub6DNfJehqF1LUs9kwaQDu12Ajzp9sYVtbGhTykQo1CYdkkqV2yAvR4DiWSTTdujWhzVy1AtV6ENGKgwLWqa4wXMR4ZmdpRjQBG5EgTV/1/*,[b8688df1/84'/0'/1']
  ↪ xpub6DNfJehqF1LUs9kwaQDu12Ajzp9sYVtbGhTykQo1CYdkkqV2yAvR4DiWSTTdujWhzVy1AtV6ENGKgwLWqa4wXMR4ZmdpRjQBG5EgTV/1/*,[b8688df1/84'/0'/1']
  ↪ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↪ xpub6DNfJehqF1LUs9kwaQDu12Ajzp9sYVtbGhTykQo1CYdkkqV2yAvR4DiWSTTdujWhzVy1AtV6ENGKgwLWqa4wXMR4ZmdpRjQBG5EgTV/1/*,[b8688df1/84'/0'/3']
  ↪ xpub6DNfJehqF1LUs9kwaQDu12Ajzp9sYVtbGhTykQo1CYdkkqV2yAvR4DiWSTTdujWhzVy1AtV6ENGKgwLWqa4wXMR4ZmdpRjQBG5EgTV/1/*,[b8688df1/84'/0'/3']
  ↪ multi_a(1,[28645006/84'/0'/0']xpub6DBbzvudcQg2nS4tHSkrm7FGkXL6arWxEyoZ1g1GWbaNdWRobcJmxH8KQdjuZTJztwDveibGd83eGzdGCR6Nj5qpU8p2Xx4wWQK7iGBycAm
  ↪ /1/*,[28645006/84'/0'/1']xpub6DBbzvudcQg2nS4tHSkrm7FGkXL6arWxEyoZ1g1GWbaNdWRobcJmxH8KQdjuZTJztwDveibGd83eGzdGCR6Nj5qpU8p2Xx4wWQK7iGBycAm
  ↪ /1/*,[28645006/84'/0'/2']xpub6DBbzvudcQg2nS4tHSkrm7FGkXL6arWxEyoZ1g1GWbaNdWRobcJmxH8KQdjuZTJztwDveibGd83eGzdGCR6Nj5qpU8p2Xx4wWQK7iGBycAm
  ↪ )#zvdvstr",
  "internal": true,
  "range": [
    0,
    999
  ],
  "timestamp": 0
}
]
note: stdout is watch-only – public keys only, cannot spend
```

But **BSMS / BIP-129** is not available for taproot (BIP-129 predates BIP-386) -- the toolkit says so and points you elsewhere:

```
$ mnemonic export-wallet --descriptor "$(cat taproot.desc)" --format bsms --network mainnet
error: --format bsms does not support taproot (P2trMulti); BIP-129 $! prerequisites do not yet include BIP-386. Real emit support is tracked at FOLLOWUP
  ↪ 'bsms-taproot-emit' and depends on a BIP-129 spec update. Use --format bitcoin-core (Core-importable) or --format sparrow (Sparrow JSON, taproot
  ↪ -capable) for taproot watch-only setup.
```

6.5 Adding a condition: which depth? (a spending-cost comparison)

Suppose you want a fifth spend path: a **new key Knew plus the preimage of a RIPEMD-160 hashlock** (secret word "please"). Should it be **folded** into an existing leaf (tree stays depth-1) or given its **own leaf** (forcing depth-2)? Decide on spending cost, not aesthetics. A taproot script-path witness costs: (satisfaction) + the **revealed leaf script** + a **control block** of 33 + 32*depth bytes. Folding many conditions into one leaf bloats *every* spend through it (you reveal the unused branches too); a deeper leaf adds one 32-byte hash per level. A witness byte weighs 1 WU = 0.25 vB, so **one extra depth level = +8 vB**.

The new hashlock (same two-step scheme): preimage $x = \text{sha256}(\text{"please"})$, descriptor hash $H_p = \text{ripemd160}(X)$:

```
$ python3 -c "import hashlib; X=hashlib.sha256(b'please').digest(); print('preimage X =', X.hex()); print('hash Hp =', hashlib.new('ripemd160', X).
  ↪ hexdigest())"
preimage X = 56ccc4dcfc96534b06fc0c08a301be24f13b491484d984953cc0dba9bbb89a
hash Hp = 06d05e2f02fb90ddf98d8cd95d806ba12b27aff4
```

mnemonic compare-cost reports per-condition witness vbytes (key-agnostic -- abstract labels A,B,... are auto-dummy-keyed). **Folded** -- the new tier joins Leaf B, so spending it reveals all of tiers 3+4+5:

```
$ mnemonic compare-cost --miniscript "or_i(and_v(v:older(65535),multi(2,A,B)),or_i(and_v(v:older(4255898),multi(1,C,D,E)),and_v(v:pk(F),ripemd160(06
  ↪ d05e2f02fb90ddf98d8cd95d806ba12b27aff4))))"
Input: or_i(and_v(v:older(65535),multi(2,A,B)),or_i(and_v(v:older(4255898),multi(1,C,D,E)),and_v(v:pk(F),ripemd160(06
  ↪ d05e2f02fb90ddf98d8cd95d806ba12b27aff4))))
Wrapper comparison: wsh(M) vs tr(NUMS, {M})
Feerate: 1.0 sat/vB
```

Condition	wsh vB	tr vB	Δ vB	wsh sats	tr sats	Δ sats
A + B + older(blocks)	79	147	+68	79	147	+68
C + older(512s)	61	132	+71	61	132	+71
D + older(512s)	61	132	+71	61	132	+71
E + older(512s)	61	132	+71	61	132	+71
F + preimage(h0)	69	139	+70	69	139	+70

note: per-condition vbytes are rounded individually; absolute numbers may differ by ±1 from real-tx accounting, Δ values are correct
 note: input contains hash-preimage fragments; preimage-known rows are enumerated assuming the user can supply each preimage (cost only – no preimage
 ↪ knowledge is implied)

Dedicated -- the new tier is its own leaf, revealing only itself:

```
$ mnemonic compare-cost --miniscript "and_v(v:pk(F),ripemd160(06d05e2f02fb90ddf98d8cd95d806ba12b27aff4))"
Input: and_v(v:pk(F),ripemd160(06d05e2f02fb90ddf98d8cd95d806ba12b27aff4))
Wrapper comparison: wsh(M) vs tr(UMS, {M})
Feerate: 1.0 sat/vB
```

Condition	wsh vB	tr vB	Δ vB	wsh sats	tr sats	Δ sats
F + preimage(h0)	68	90	+22	68	90	+22

note: per-condition vbytes are rounded individually; absolute numbers may differ by ± 1 from real-tx accounting, Δ values are correct
note: input contains hash-preimage fragments; preimage-known rows are enumerated assuming the user can supply each preimage (cost only – no preimage
↳ knowledge is implied)

compare-cost models each input as a single leaf at the tree root (depth-0, 33-byte control block); add the real Merkle depth (+8 vB per level):

Placement	revealed script	tr vB (depth-0)	real depth	+ control block	real tr
Folded into	whole	139	1 (still 2	+8	~147 vB
Leaf B	or_i(t3,t4,t5)		leaves)		
Dedicated leaf	just t5	90	2 (now 3	+16	~106 vB
			leaves)		

A dedicated leaf is ~41 vB (~28%) cheaper to spend the new condition: not revealing tiers 3+4 (~49 vB) beats the +8 vB for the extra depth level. It is better still in practice because folding also makes *t3* and *t4* spends reveal *t5*'s script, and a 3-leaf tree can place the **most-used** condition in the shallow slot and bury cold paths deeper.

Rule of thumb: in taproot, one-condition-per-leaf almost always wins -- each sibling script you avoid revealing is worth far more than the 8 vB/level of depth -- and you order leaves hot-shallow / cold-deep.

The catch: cost says depth-2, but master cannot build depth- ≥ 2 yet (the PR-#953 taptree-Display bug, 6.1), so today the only buildable option is the folded depth-1 leaf at the ~28% premium. That premium is the concrete, quantified motive for landing the upstream fix.

6.6 Taproot multisig (NUMS), cross-checked against Bitcoin Core

The taproot wallets above are deliberately complex. The *simplest* taproot **multisig** is a single `sortedmulti_a` 2-of-3 leaf under the BIP-341 **NUMS** ("nothing-up-my-sleeve") internal key -- an **unspendable** key-path, so the only way to spend is the sorted multisig script. It round-trips on the **shipped** binary, and Bitcoin Core derives the **identical** first address. (The toolkit's own `tests/bitcoind_differential.rs` gate proves exactly this -- `bundle -> restore -> first_addresses` VS `Core deriveaddresses` -- for the taproot-multisig corpus against a pinned Bitcoin Core v27.0.) Reuse the three section-3 cosigners:

```
$ cat taproot-multi.desc
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,sortedmulti_a(2,[73c5da0a/87'/0'/0']
  ↳ xpub60BjiYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub6CbhrPzY2z7NcCGGjLAJLq8iRyJufwmdXQs66MxTVURekQb9DpLnVJ5D1qpatZjUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokYLVsyEFZFt9/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub6DB7HNqw6CZojxN85NuFTPWZhi2FagSnextPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z/<0;1>/*))
```

Validate and canonicalise it (the NUMS hex is the BIP-341 unspendable H-point):

```
$ mnemonic export-wallet --descriptor "$(cat taproot-multi.desc)" --format descriptor --network mainnet
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,sortedmulti_a(2,[73c5da0a/87'/0'/0']
  ↳ xpub60BjiYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub6CbhrPzY2z7NcCGGjLAJLq8iRyJufwmdXQs66MxTVURekQb9DpLnVJ5D1qpatZjUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokYLVsyEFZFt9/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub6DB7HNqw6CZojxN85NuFTPWZhi2FagSnextPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z/<0;1>/*))#8nz0lwja
note: stdout is watch-only – public keys only, cannot spend
```

Engrave the watch-only card set and read the first address back from the md1 chunks alone:

```
$ mnemonic bundle --descriptor-file taproot-multi.desc --network mainnet --json | jq -r ".md1[]" > taproot-multi.md1
note: stdout is watch-only – public keys only, cannot spend
```

```
$ mnemonic restore --network mainnet $(sed "s/^/--md1 /" taproot-multi.md1)
2-of-3 multisig restore
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,sortedmulti_a(2,[73c5da0a/87'/0'/0']
  ↳ xpub661MyMwAQbRcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzwBTBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPwDUSNJsY5JsgVBQoD6KrbdcLL/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub661MyMwAQbRcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXYukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmrEkfhjzgHCUEcvJBhXS5bk4RW/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub661MyMwAQbRcEdy4jr5EtEhQBctfscE6a99DGLr2cW4HnnmBsXDoE3odGzRiw3hCRMSwfkCQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/<0;1>/*))#k0lsap8u
first recv: bc1p550zvnachy40z6hh8llka93mkmc3635sampp264ck6rfd0dcd8s00n8c8
cosigner @0: 73c5da0a [87'/0'/0'] from md1 (not independently verified)
cosigner @1: b8688df1 [87'/0'/0'] from md1 (not independently verified)
cosigner @2: 28645006 [87'/0'/0'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only – public keys only, cannot spend
```

`restore` reports a `bc1p...` Taproot address. Confirm it against Bitcoin Core's **independent C++** derivation: `deriveaddresses` on the receive (`.../0/*`) descriptor (split from the `<0;1>` multipath, which Core rejects):

(STATIC CAPTURE -- recorded from Bitcoin Core v27.0. `deriveaddresses` is a deterministic descriptor-to-address function of the fixed descriptor above; this line is NOT regenerated by `gen.sh` and needs no node in CI.)

```
$ bitcoin-cli -chain=main deriveaddresses "tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,sortedmulti_a(2,[73c5da0a/87'/0'/0']
  ↳ xpub661MyMwAQbRcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzwBTBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPwDUSNJsY5JsgVBQoD6KrbdcLL/0/*,[b8688df1/87'/0'/0']
  ↳ xpub661MyMwAQbRcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXYukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmrEkfhjzgHCUEcvJBhXS5bk4RW/0/*,[28645006/87'/0'/0']
  ↳ xpub661MyMwAQbRcEdy4jr5EtEhQBctfscE6a99DGLr2cW4HnnmBsXDoE3odGzRiw3hCRMSwfkCQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/0/*))#mk8vdqmt" "[0,0]"
["bc1p550zvnachy40z6hh8llka93mkmc3635sampp264ck6rfd0dcd8s00n8c8"]
```

Byte-for-byte the same `bc1p...` that `restore` reported -- the toolkit's own derivation (which v0.49.1 routes *around* the codec for taproot) agrees with Bitcoin Core. `tr(NUMS,sortedmulti_a)` is a shape **only** the toolkit can render -- its pinned rust-miniscript fork has `sortedmulti_a`, the codec's `crates.io` build does not -- so this end-to-end check is the only place an external oracle sees it.

Appendix A -- the public test seeds used

Slot	BIP-39 test phrase	Fingerprint (BIP-87)
@0	abandon abandon abandon abandon abandon abandon ↳ abandon abandon abandon abandon abandon about	73c5da0a
@1	legal winner thank year wave sausage worth ↳ useful legal winner thank yellow	b8688df1
@2	letter advice cage absurd amount doctor acoustic ↳ avoid letter advice cage above	28645006

These are world-known BIP-39 vectors with no funds. Sections 5-6 derive their keys from these same three seeds at distinct `m/84'/0'/N'` accounts. Generated with `mnemonic v0.85.0` on Linux. See the in-repo manual (`docs/manual/`) for the authoritative per-flag reference.

Appendix B -- EXPERIMENTAL: depth->=2 taproot reconstruction (NOT FOR REAL FUNDS)

EXPERIMENTAL -- DO NOT USE FOR REAL FUNDS. Everything in this appendix uses `mnemonic` → `-depth2`, a proof-of-concept binary built from the never-merged branch `experimental/taproot-depth-ge2`. It pins an **unreleased** rust-miniscript commit (the PR-#953 merge, in no crates.io release) to lift the `depth->=2` taptree cap. It is **not** part of any shipped release, is not on the install path, and must never be used to secure real funds. The shipped `mnemonic` v0.85.0 documented everywhere else in this guide deliberately **refuses** `depth->=2` (sections 6.1 and below); that refusal is the supported behavior until a rust-miniscript release > 13.1.0 ships #953.

Master refuses to *restore* a `depth->=2` taptree because the shipped miniscript pin mis-formats nested taptrees. The POC bumps that pin and lifts the cap. Build it under a **distinct name** so it never overwrites your real `mnemonic`. The twin below was built from the 0.55.3-era experimental branch; the static reconstruction capture further down is from that build:

```
$ git clone https://github.com/bg02h/mnemonic-toolkit && cd mnemonic-toolkit
$ git checkout experimental/taproot-depth-ge2
$ cargo build --release --manifest-path crates/mnemonic-toolkit/Cargo.toml
$ cp target/release/mnemonic ~/.cargo/bin/mnemonic-depth2 # DISTINCT name
```

Recall the `depth-2` four-leaf descriptor from section 6.1 (one tier per leaf):

```
$ cat taproot-4leaf.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNTNqr2UHCBEsCol3eEFvAM7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLix6XZs6XQmGcQNe04HcjfVTn9TuXE/<0;1>/*,{and_v(v:
  ↳ after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↳ xpub6CatWdiZiodmUeTDp8LT5or8nmbKNcuyvz7WysVfKB4RHwCD3XyuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↳ xpub6CatWdiZiodmYVtWLTtEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcycJgEzV3quZSJKASdh5Lo2PbubMGxCfZtRthV6ST2qquL9w3HSzcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↳ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNNi9WriYgyf7VxcmuFxsosRw3focZLgRdbjDjJbqZPCTfKcaWmL9BuSw98ybKKJtcHgWerry6/<0;1>/*)),and_v(v:after
  ↳ (1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↳ xpub6CatWdiZiodmHurRokjbycCrxdTDJgTsyEaaQfKjKwBuUi79LAWG5GhJMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVdLdPkEC/<0;1>/*,[b8688df1/84'/0'/0']
  ↳ xpub6DNfJehqFILLUs9kwaQDu12Ajzp29sYVtbGhTykQo1CYdkkqV2yAvR4DiWSTTDUjWhzVy1AtV6ENGKwGwLWqa4wXMZR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↳ xpub6DNfJehqFILLUsh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwY71Nc6LvkqFDLWuyCivJ4eMDXgFtiTknazgp8rYvZQSKR3L8EGRs9asBarqpe/<0;1>/*)),{and_v(v:older
  ↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↳ xpub6DNfJehqFILLUwZ1DrFcQN365CDMPLSsRpSwaz4eRrtYz9qWdsVH9JqsHp06yNGdG4VXGKbLgcXKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↳ xpub6DNfJehqFILLUxb8gPsStbR9LcAxhgwD4V678z5FZ7BLftWzwQct3yZb5eW4U8AVJwbKcZ2ieggjvkV2xggjPMBRkk3CE7bz7g4uMV7Qp3TU/<0;1>/*)),and_v(v:older(4255898)
  ↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvudcQg2nS4tH5Krm7F6KXL6arWxYoZ1g1GwbaNdWRobCJmxH8KQdjuzTzJtwDveibGd83eGzdgCR6NjSqpu8p2Xx4wWQK7iGBycAm
  ↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvudcQg2sPDRWpqtEn5VzPiwr1zNes4aHmnuOug9Jmc2fc2JSM2E39Yzy2iakmpwRpa3rXhzGNmd4GK1JEZxmTCftQwBgGd9ihVks
  ↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvudcQg2uk9BrsxusCWjSsYsfPYkPahQfmTVABFJ4j8TRxUnRd5eGgEXfJP63xcuTV9uny7pQBFbu3XKCNr9NxcNGRaDT9BD1gkBGZ
  ↳ /<0;1>/*)))
```

The engraved card is a faithful backup even on master -- `bundle` accepts it; only *reading it back* is gated. Make the card and pull its `md1` chunks with the shipped binary:

```
$ mnemonic bundle --descriptor-file taproot-4leaf.desc --network mainnet --json | jq -r ".md1[]" > depth2.md1
note: stdout is watch-only – public keys only, cannot spend
```

```
$ cat depth2.md1
md1f8eucvq9zem6jzwrh4yypm6jzxw75sj3m6jzt802ggrh4yyvaafp9rh4yykw75s802ggemcsk6ef96dmuhe
md1f8eucvq2jz2sqrqgY29fntvqq7sjgqntk5ymnjqatj0sucqg70h4gdtkemj3r4wfp6rc6cqzhan8ac8kjjyw
md1f8eucvqscckmmq5mmyg26g2v66mwrda3qzv6a4pxuusyht2unu8xqz8naa2r2akwukvgm4gaa2pacrd4qv84
md1f8eucvq6gws7xkx9k7c9xu6pzkjg9jv9fntsqpllllyqshsnxhqggrcf5gqzn2cvasqu79capl4rdda7hr0z
md1f8eucvprg9pw0za5z3883w6pgmncdq53euts2twrg3hckhp5gmuts6ydx9cdzx1ry5xg4vfvrdrp8fdd6
md1f8eucvpg5q52ry2rt9p9j9pqsufq44hccjz40ltn8tdswlxxk6gs50wsfd6vu4jev3egrh7l0chmf9x6
md1f8eucvp56gl4vs3t2qswfth67hsept4g4hf5939sedkqhey4nnc0chj6t2x82xggygmkvatgel8qeexkxd
md1f8eucvp622ws2kgdemj2cd8zwj2skzx2wq0w70l4q99vdyh5x0z8v4yslsp8q7nzlkkvymsohv0v5afqaykz
md1f8eucvz9fkac3cxteqm6kacqcp59u7xtux3d5d847wsf4xev6setz7wcgwg0mnt0jaragegr6d8x0glwkqkw
md1f8eucvz0ptr0vxf3c9895ghka0gk58rcw6qzw4cdxd3en9mnxhdpqws2ec6jxdf33axqcausm7pgz4f8s
md1f8eucvznmj3skn4t8pc6l88xna4nqjquhu832y6rwxxt8b3xuf33474jxxpl30507ttsmqyxtz2nkucsqsf
md1f8eucvzcxc7rcsyrzx48lzn7wtu6t526tvfexawdfdy4vy7y7s9c84gwagyhfh8ugmwft46ekq8ppdgxrw6l86
md1f8eucvrr0a7t0c6gn7uss8mxrkma6e6mkap24zvzxnc1w0xxzg0sgsqtmmwy87px3u9mlm3vgw5yc5xfg4v5q3
md1f8eucvrvnm3jaafe7gyny7hjgysfhfz4faf5hs5jwzh3mhpgjm2ct2uevmwdfap2f5z19lsrnz16nax3u2eu
md1f8eucvprkdtfh372tg8y3r2d35qm06q30450k4vmqvakpyw7pdkdanh95g2a605szqpgggn3cjkltniejh6
md1f8eucvrgz55fzuu4z339d06w5s99f4v94tc38peqcv5jyspp3g08exa4srx0889h3ddqsnr60kjlukyy4
md1f8eucvyp6tmtcl5y7ac6vels0u3jj86dd5pe7m355rtw8r3a2s45tsczmremst4u6j4atqmt5zth78a5h00
md1f8eucvvgu6h0m3m5l1v9u9cde6pcswdr73ejs4lts9qcszq6fnt9u8jvs93uep30g6dqqv99qzfcynrx60
md1f8eucvynng0aj0c55p20eykxhftlk456p2lvcqh3za8msktlmcqvqkmuj2dwpkhhlhm290qc60psjjszcz7v2a
md1f8eucvykdg4nrpzxs5z4jyqut3ee64sfnz6ryr348l5ljx4sv0e02c2fml4zmy9g36cer4vv2u2lmgk4
md1f8eucv9rfgzjjzpmz7nhnk6yhsveeegnzkfxcgdg3sypfknoR6gc370szpx6rkdq27exceqsqdl8x62tamz
md1f8eucv9tzzetsg6tpqua7cty0t4c4w6rtmjnmht8891lvhne3ktx5etuhl2rmxpaqj6ey0lge7pv3gpppm09
md1f8eucv95xj46th52v3c3kj4u0767f9u9j0ynrejo0uqenhu2pdakxrtf5e5qck40z57z8ylnqr9yt5gn9a5u
md1f8eucv9eyht2qvnhxckz5khfacj7d2lxxnax2ywfue5grx4rgemhg0nk5eg0ze8tj0qfnr3s8qltzh5k7fvar
md1f8eucvx92k0c250h6ypsarcmrl7tgutcg8uf4m7gfc8lrv33knprwhtl06am5gdw9aqpfnr9h10fq5pp9p
```


The shipped `mnemonic` refuses to restore it (the supported behavior -- the card stays a faithful backup):

```
$ mnemonic restore --network mainnet $(sed "s/^/--md1 /" depth2.md1)
error: taproot tree depth ≥ 2 ≥ (3 leaves) is not yet restorable – the pinned miniscript mis-prints nested taptrees (FOLLOWUP upstream-miniscript-taptree-
↳ depth2-display-asymmetry); the engraved card remains a faithful backup
```

The experimental `mnemonic-depth2` reconstructs the very same card -- depth-2 taptree and all -- and prints a loud EXPERIMENTAL advisory on stderr before the reconstructed descriptor and first address:

(STATIC CAPTURE -- recorded 2026-06-15 from the experimental `mnemonic-depth2` build at v0.55.3. This block is NOT regenerated by `gen.sh` and is not reproducible with a released binary or in CI; build the branch as shown above to reproduce it. Everything else in this document is live-captured and CI-gated.)

```
$ mnemonic-depth2 restore --network mainnet $(sed "s/^/--md1 /" depth2.md1)
EXPERIMENTAL: depth≥2 taproot reconstruction relies on an UNRELEASED rust-miniscript commit (#953, in no crates.io release) – proof-of-concept only; do
↳ NOT use for real funds and do NOT merge. Rebuild when miniscript > 13.1.0 ships.
miniscript policy restore (12 cosigners)
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: tr([73c5da0a/84'/0'/4']xpub661MyMwAQbRcEYUKSqsBgazlLob8pCa1rM1Sj8CEzGCYyP9LisxZ2mIgoDqj137XvHdY2nNkctqiElixaFqYHf9lCFpFpKicVb7TzvrGsE
↳ /<0;1>/*,{&and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0flac62dbd829b9a08ad),multi_a(3,[73c5da0a
↳ /84'/0'/0']xpub661MyMwAQbRcFHMVYpCiBTXdtCaj7vZnNFHJ5g59Aue2yYkSrZ5q9Gaq4rRjJVhHtZtCiHWSzgmS5beaaTHWVmhpGC7SMdqMXHRXZi8as/<0;1>/*,[73c5da0a
↳ /84'/0'/1']xpub661MyMwAQbRcGaxoYcLaxHHXZqEgSRQmN2P5ung8M38MNE535mLuhq7zjnrMkyA5eX6ehicVbU1FFPU39LgXbY8PmPLQxVRQmPfa3Q7spa/<0;1>/*,[73c5da0a
↳ /84'/0'/2']xpub661MyMwAQbRcGuXAHBK3oquZS1HJiz2fVZ2idNcK4GLGTJyGZkPK7fviN6euv5GzY18JD3WBG3SoLat23TLAVjhQmXDVMAqymQNhg3RFT8/<0;1>/*)),and_v(v:
↳ after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0flac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
↳ xpub661MyMwAQbRcGHLcZcLjg25oG8wSyqSE5XNM9uMks6vrpH4pRDC8UmAynovThuKraidMeEKJ2FcqBwleF76aeu1vrGtLJXUijXr4r9NITZQ/<0;1>/*,[b8688df1/84'/0'/0']
↳ xpub661MyMwAQbRcGowNgeNcLS8Cg12vnZybpJqtkbCmSQMdq2qzCDWq3CXG7x5BqvcNCsANUw6nisoN7JFK2j3HfxV57nNm2RLKo2UzHgbs6/<0;1>/*,[b8688df1/84'/0'/1']
↳ xpub661MyMwAQbRcEv3U8uuxavsQA8LNNYwcNge8rT7SaMS558KiEwXoP72TQ8ARYjczPtTQz6CxcabTEE3XchmYvSiHcVbC9h17CmyfG7sVq9/<0;1>/*))),{&and_v(v:older
↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
↳ xpub661MyMwAQbRcG7Xht9EwgNucA47Rmgg8Bn5bNmFJMkothQDXirpogQHkVNRcAy6GkGnUYUMNBFcNaRq4WnsqW2VNUddW6ymHXfVpK4c3/<0;1>/*,[b8688df1/84'/0'/3']
↳ xpub661MyMwAQbRcEbqBvNkLuDtudGA2PHAbtWUuHKe3CKjZCaLjxLGS68SjpwBCnXsj8xPGXaV9ZWL3j9ktbed8y1aeNVK95HrkgHfHG8XM5Eh/<0;1>/*)),and_v(v:older(4255898)
↳ ,multi_a(1,[28645006/84'/0'/0']xpub661MyMwAQbRcEdBofBaGbgns74WRuyEbXRSzmzq8jzthztDnXTV2yNQPzgs3ubwuNp7yrSHnECoa5xHgnoEDH4HSGWqLtYdi6nWVZCfXPK
↳ /<0;1>/*,[28645006/84'/0'/1']xpub661MyMwAQbRcH2WNMBtz4pZ8wDtpxndYo6E4r5o8pXedve17srmalLCEjM8WcpV67xsc36KpBntYUdqo5dpcFMzRfzSSa4C5DRty4eDNF
↳ /<0;1>/*,[28645006/84'/0'/2']xpub661MyMwAQbRcGqcAAnB9mhvQsdUx2fKasUoXt2Gmpt2tFz94wRfAkhuLhZUJkQ5pgnd9Ny9EwrgcHbAASVnQShCbfnGskAk2k6yGoWXAv
↳ /<0;1>/*))))#5trrrgdg0
first recv: bc1p6yc7kzttsafpr6hwsaefuyqxvee4j48zdrqt4kl9ers68mhcestwvnn66
cosigner @0: 73c5da0a [84'/0'/4'] from md1 (not independently verified)
cosigner @1: 73c5da0a [84'/0'/0'] from md1 (not independently verified)
cosigner @2: 73c5da0a [84'/0'/1'] from md1 (not independently verified)
cosigner @3: 73c5da0a [84'/0'/2'] from md1 (not independently verified)
cosigner @4: 73c5da0a [84'/0'/3'] from md1 (not independently verified)
cosigner @5: b8688df1 [84'/0'/0'] from md1 (not independently verified)
cosigner @6: b8688df1 [84'/0'/1'] from md1 (not independently verified)
cosigner @7: b8688df1 [84'/0'/2'] from md1 (not independently verified)
cosigner @8: b8688df1 [84'/0'/3'] from md1 (not independently verified)
cosigner @9: 28645006 [84'/0'/0'] from md1 (not independently verified)
cosigner @10: 28645006 [84'/0'/1'] from md1 (not independently verified)
cosigner @11: 28645006 [84'/0'/2'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only – public keys only, cannot spend
```

The reconstructed descriptor is the genuine depth-2 shape `tr(Kint,{{A,B},{C,D}})` (four leaves, one tier each) -- the layout section 6.5 showed is cheaper to spend but that master cannot yet build. When a rust-miniscript release > 13.1.0 ships #953, this is rebuilt as a supported feature and this appendix retires.